gain
Global Alliance for
Improved Nutrition

Training Manual for Vitamin A Cassava

Raw Material Quality, Handling Finished Product, Packaging and Storage to Meet Quality and Regulatory Standards in Nigeria

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FOREWORD

Detailed overview of GAIN and the SNIP project.

DISCLAIMER

ACKNOWLEDGMENTS

This training manual was produced under the Strengthening Nutrition in Priority Staples Project (SNIP), implemented by the Global Alliance for Improved Nutrition (GAIN), in partnership with GIZ and the Green Innovation Centre for the Agricultural and Food Sector in Nigeria (GIC), with support from the German Government through the Federal Ministry for Economic Cooperation and Development (BMZ).

SNIP aims to strengthen the GIC priority value chains and improve food security and safety through improved diets for farmers, farm workers, other workers in businesses in these value chains, and among the wider population in Nigeria.

This training manual was developed by a team of consultants from Citadel Premium Bridge Limited, with project oversight led by Mr. Mosi Akande, technical oversight by Mr. Feyisara Oguranti and Ms. May Kachukwu Onwodi, and technical reviews by Dr. Funke Olotu (Ph. D). The consulting team would like to thank all personnel of the GAIN for their dedication and assistance throughout this training project, and particularly to Mr. Godwin Ehiaibhi, Ms. Mercy Olorunfemi, and Mr. Orngu Africa, for their consistent leadership and support.

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TERMS, ABBREVIATIONS, AND DEFINTIONS

1. Carotenoid:

A class of pigments found in plants, algae, and some microorganisms. Carotenoids are responsible for the red, orange, and yellow colors in fruits and vegetables. They also act as antioxidants and may have health benefits when consumed as part of a balanced diet.

2. FIFO (First-In, First-Out):

A method of inventory management and product rotation where the oldest items in stock are used or sold first. This ensures that products are used before they expire or become obsolete, minimizing waste and maintaining product freshness.

3. EEFO (Earliest Expiry, First Out):

A variant of the FIFO method specifically used in inventory management, where the items with the earliest expiration dates are used or sold first. This approach helps prevent product spoilage and ensures that items with shorter shelf lives are consumed before those with longer shelf lives.

5. IPM (Integrated Pest Management):

An approach to pest control that focuses on using a combination of preventive measures, monitoring, and targeted interventions to minimize pest populations and damage to crops. IPM aims to reduce reliance on chemical pesticides and promote sustainable pest management practices.

6. Infestation:

The presence or invasion of pests, such as insects, rodents, or microorganisms, in large numbers or quantities that can cause harm, damage, or contamination to crops, stored food, or premises.

7. Deficiencies:

In the context of nutrition, deficiencies refer to insufficient intake or inadequate absorption of essential nutrients, vitamins, or minerals required for optimal health. Nutrient deficiencies can lead to specific health problems or conditions.

8. Cross Contamination:

The transfer of harmful microorganisms, allergens, or contaminants from one surface, food item, or area to another. Cross contamination can occur during food preparation, storage, or handling and can compromise food safety.

9. Food Fraud:

The deliberate and deceptive practice of substituting, adulterating, mislabeling, or misrepresenting food products for economic gain. Food fraud involves fraudulent activities that can impact consumer safety, quality, and authenticity of food items.

10. Inventory:

The complete list or record of goods, products, or materials available in stock within a business. Inventory includes information about quantities, locations, and other relevant details for effective management and control.

11. Bincards:

Physical or digital cards that provide detailed information about specific items or products in inventory. Bincards typically include data such as item descriptions, quantities, locations, and transaction history.

12. Batches:

A group or quantity of products or materials produced or processed together, often with the same characteristics, specifications, or production dates. Batches are used for tracking and quality control purposes.

13. Pallets:

Flat platforms or structures made of wood, plastic, or metal that are used for storing, stacking, and transporting goods. Pallets provide stability and convenience in handling and moving products, especially in warehousing and logistics operations.

14. Pest:

Any organism, typically insects, rodents, birds, or microorganisms, that can cause damage, contamination, or harm to crops, stored food, structures, or human health.

15. Corrective Control:

Measures or actions taken to address deviations, non-conformities, or issues identified during quality control or food safety processes. Corrective control aims to identify and rectify the root cause of a problem to prevent its recurrence.

16. Root Cause Analysis:

A systematic approach used to identify the underlying or fundamental cause of a problem, incident, or non-conformity. Root cause analysis helps determine why a particular issue occurred to develop effective corrective and preventive actions.

17. Garri:

Garri is a popular West African food made from cassava roots. It is typically processed by peeling, washing, fermenting, and grinding the cassava, followed by drying and sieving to produce granules or flour. Garri is commonly used as a staple food and can be prepared in various ways.

18. CIP (Clean-in-Place):

A cleaning method used in food and beverage processing industries to clean equipment, pipes, and tanks without disassembling them. CIP involves circulating cleaning solutions and detergents through the equipment to remove residue, bacteria, and other contaminants.

19. REORDER LEVEL:

The predetermined inventory level at which it is necessary to reorder or restock a particular item or product. Reorder level ensures that sufficient stock is maintained to meet demand and prevent stockouts.

20. Outsourcing:

The practice of contracting or delegating certain tasks, functions, or operations to external companies or service providers. Outsourcing allows businesses to access specialized expertise, reduce costs, and focus on core activities.

21. Criteria:

Specific standards, requirements, or conditions that must be met to evaluate or assess the quality, suitability, or acceptability of a product, process, or service.

22. Food Grade:

Refers to materials, ingredients, or products that meet the necessary quality and safety standards for use in food processing, packaging, and handling. Food-grade materials must not pose any risk of contamination or harm to consumers.

23. SOP (Standard Operating Procedure):

A documented set of step-by-step instructions or guidelines that outline the standard and approved methods for carrying out specific tasks or processes. SOPs are used to ensure consistency, quality, and safety in operations.

24. Food Handlers Test:

A test or examination conducted to assess the knowledge, understanding, and competency of individuals who handle or work with food. Food handlers tests aim to verify that individuals have the necessary understanding of food safety practices and regulations.

25. PPE (Personal Protective Equipment):

Protective clothing, equipment, or gear worn by individuals to minimize the risk of injury, illness, or contamination in the workplace. Examples of PPE in the food industry include gloves, aprons, goggles, and masks.

26. NAFDAC (National Agency for Food and Drug Administration and Control):

NAFDAC is a regulatory agency in Nigeria responsible for regulating and controlling the manufacture, importation, exportation, distribution, advertisement, and use of food, drugs, cosmetics, medical devices, and chemicals.

27. SON (Standards Organisation of Nigeria):

SON is a statutory body in Nigeria responsible for establishing and enforcing standards for products, processes, and services to promote consumer safety, quality, and fair-trade practices.

28. NQS (National Quarantine Service):

The National Quarantine Service is a regulatory agency that oversees the inspection, certification, and quarantine of agricultural commodities, plants, and other related materials to prevent the introduction and spread of pests, diseases, or contaminants.

29. IPAN (Institute of Public Analyst of Nigeria):

IPAN is the institute that regulates the practice of Laboratory analyst as well as the release and monitoring results for third party analysis in Nigeria

30. Third Party/Comprehensive Testing:

Testing conducted by an independent third-party laboratory or testing facility to assess the quality, safety, composition, or compliance of products or materials. Comprehensive testing provides objective and unbiased results.

31. Hermetic:

Airtight or sealed tightly to prevent the entry or escape of air or moisture. In the context of food storage, hermetic containers or packaging help maintain product freshness and prevent spoilage.

32. MANCAP (Mandatory Conformity Assessment Program):

MANCAP is a certification program in Nigeria that verifies and certifies the conformity of locally manufactured products with relevant standards and specifications.

33. Primary Packaging:

The initial layer or container that directly holds a product. Primary packaging is in direct contact with the product and is designed to protect, preserve, and provide information about the product.

34. Secondary Packaging:

Packaging materials or containers used to group, protect, and transport multiple units or primary packages of a product. Secondary packaging provides additional protection, convenience, and branding.

35. Food Recall:

The action of removing or recalling a food product from the market or consumer use due to safety concerns, quality issues, contamination, or other risks. Food recalls are initiated to protect consumer health and safety.

1. INTRODUCTION AND PRODUCT BACKGROUND

Biofortified vitamin A cassava refers to a variety of cassava that has been genetically modified to contain higher levels of beta-carotene, which is a precursor to vitamin A. This biofortification process aims to address vitamin A deficiency, a significant public health issue in many developing countries where cassava is a staple food.

1.1. PRODUCTS IN THE STATE

While there are several agricultural commodities and product grown in the state, biofortified, vitamin A cassava is the focus of this training manual.

1.2. WHAT MAKES THE PRODUCT DIFFERENT

While there are several agricultural commodities and product grown in the state, Vitamin A cassava is the focus of this training manual.

1.3. DERIVATIVES FROM THE PRODUCT

A growing number of derivatives are available from vitamin A cassava. In Nigeria, some include:

- Vitamin A Cassava flour
- Vitamin A Garri
- Vitamin A fufu
- Vitamin A Cassava Chips and snacks
- Boiled Cassava products

1.4. PECULIAR CHARACTERISTICS OF THE PRODUCTS

Commonly grown cassava does not contain vitamin A. Biofortified vitamin A cassava varieties contain as much as 120 micrograms (ug) of provitamin A carotenoids per 100 grams (g) of cassava; the additional carotenoids are accumulated in the edible parts of the plant as part of the normal growing process. It is gluten free, suitable for gluten-free diets, and does not contain gliadin or glutenin.

Vitamin A cassava is naturally colorful. The vibrant yellow color is attractive to consumers. Health education campaigns urge consumers to eat a varied, colorful diet. It contains antioxidants rich in beta-carotene, an antioxidant that supports eye health.

Nutrient	Vitamin A Cassava	Commonly Grown Cassava
Carbohydrate	32g/100g	32g/100g
Protein	1.9g/100g	1.9g/100g
Fat	0.8/100g	0.8/100g
Moisture	64g/100g	64g/100g

Vitamin A µg	120µ/100g	0

2. POST-HARVEST HANDLING

MODULE 1: HANDLING OF PRODUCT AS COMMODITY

2.1. WATER ACTIVITY

Water activity is a measurement of the availability of water for biological reactions. It determines the ability of micro-organisms to grow. If water activity decreases, micro-organisms with the ability to grow will also decrease.

Water activity in farm produce is a factor that leads to deterioration due to microbial activities. The lower water activity guarantees longer shelf life for a product. Drying for cereals reduce the water activity and improve the shelf life if properly handled.

Ensuring that product being moved for processing has reduced water activity to ensure better condition before getting to production. Reduce exposure of produce to heat and sunlight to reduce the effect of water activity. Vitamin A cassava with bruises or broken should be avoided as much as possible as the exposure can lead to easy deterioration.

2.2. SORTING AND SEPERATION

All products are to be sorted at the farm gate according to requirement of needed raw materials. Tubers can be sorted into grades depending on size and status. Vitamin A cassava can be sorted based on grain length, moisture content and the weight per bag.

Sorting helps to also separate defective products from wholesome products. Sorting helps to get only products needed and pay the right fare. Sorting for farm produce should be done based on predetermined raw materials by the producer.

All separated products must be clearly labeled with the right codes identifiable at the factory gate.

2.3. BAGGING AND PACKAGING

Farm produce storage is important as it affects the rate of deterioration of products before they get to production points.

Tubers are sometimes packed into the transport vehicles without any packaging material leaving them exposed to the environment. Wholesome tubers can get bruised over time in this condition, but this process is acceptable if the distance is considerably short, and the raw material will be used promptly.

IMAGE OF SUGGESTED BAGS AND PACKAGING MATERIALS FOR VIT A CASSAVA PRODUCE TO BE ADDED HERE

2.4. TRANSPORTATION REQUIREMENTS

Vehicles to be used to transport food products should be clean and not capable of contaminating the product they are shipping. Vehicles for transportation should be used for transporting food products only. Other products should not be packed together with the food products during transportation.

Animals should not be packed in vehicles to be used to transport farm produce. Sharp edges should be avoided in vehicles as they can damage the bagging materials. Vehicles should be in good condition and fit for the journey as braking down of vehicle can increase transport time and lead to produce deterioration.

Loading personnel should use the right protective clothing when loading the vehicles. Loading personnel should ensure they observe necessary hygiene requirements such as washing of hands. Loading personnel should ensure they arrange produce with care to avoid breakage, bruises as well as baggage damage.

2.5. IMPURITY AND CROSS CONTAMINATION

Impurity in farm produce is any other material or produce different from the primary produce that is found inside the bulk of the primary produce. Impurity may include chaffs, other grains, waste materials, fecal material, chemical materials as well as stones and sand. Impurity can lead to cross contamination of the produce from another source.

Common sources of cross contamination are animal fecal matter, agrochemicals, vehicle fuel as well as other materials which gets into the Produce unintended. Produce should be packaged and transported in a way to exclude any source of contamination.

2.6. FOOD FRAUD AND IMPLICATION

Food Fraud is an intentional activity by a supplier or vendor to tamper with the contents of produce. Food fraud may include intentionally to add stones to increase weight mixing variety of orange flesh sweet potato that is not vitamin A variety and hiding it from the buyer. For example, adding chaff to the orange flesh sweet potato bag to make it bulky and paid for in full. Food Fraud is very common with producers in Nigeria and unsuspecting buyers usually fall victim.

2.6.1. FOOD FRAUD IMPLICATION

- Less Value for money and reduced profit margin for producers
- Reduces product quality
- Increases required production activities
- May cause damage to machinery
- Can affect company reputation
- May disrupt a complete production schedule

2.7. AVOIDING FOOD FRAUD

- Ensure an expert based on education, experience or training is checking produce before, during and after transportation
- Have a complete supplier program where suppliers are preselected and worked with based on set criteria
- Have a complete out-growers system where the farm input and output are controlled
- Carry out Quality Assurance on produce at point of purchase using simple hand-held testers
- Buy produce from farm associations and cooperatives who are more accountable
- Appoint middlemen who take the risk of purchase and ensure proper quality assurance at the factory gate

3. RAW MATERIAL QUALITY AND HANDLING

MODULE 1: INVENTORY AND MATERIAL BALANCE

3.1. RAW MATERIAL INVENTORY: FIFO AND EEFO PRINCIPLES

- First In-First Out (FIFO) principle helps to ensure that raw materials are used in the order they were received. A raw material received later should not be used before the ones received earlier.
- Early Expiry First Out (EEFO) principle advises that products should be used in the order of expiry. Products that will expire first should be used first. The principle of EEFO overrides FIFO if it needs to be applied.

- EEFO should be used when physical defect is involved, although subjective, experienced personnel can determine which product should be used first based on existing defects.

3.2. RAW MATERIAL RECORD KEEPING: USE OF BIN CARDS

- Label each incoming batch correctly including name of supplier, quantity, who accepted the batch. Attach bin cards on each batch.
- Monitor that FIFO and EEFO are followed based on Bin cards
- Ensure the bin cards are filled appropriately when raw materials are taken out of the store.
- Where bin cards cannot be used, proper labeling of batches should be done to identify the batch, the receiving authority as well as when it was received and if possible, condition it was received. This process helps to also identify batches as well as monitor usage. It also helps to keep track of reorder level and monitor rate of reorder.

TEMPLATE OF BIN CARDS TO BE ADDED HERE

3.3. LABELLING OF BATCHES

Proper labeling of batches is crucial for the storage of cassava to ensure traceability and facilitate effective inventory management. Here are some guidelines for labeling cassava batches:

- **Batch Identification:** Assign a unique identifier or batch number to each batch of Vitamin A Cassava. This number can be alphanumeric and should be easily distinguishable.
- **Date of Harvest:** Clearly indicate the date of harvest on the label. This helps in determining the freshness and shelf life of the Vitamin A Cassava.
- **Variety:** Specify the variety of Vitamin A Cassava on the label. Different varieties may have distinct characteristics and storage requirements, so it's important to differentiate them.
- **Farm or Source:** Include the name or code of the farm or source from where the Vitamin A Cassava were harvested. This information helps with traceability and enables identification of the origin of the produce.
- **Storage Conditions:** Note the recommended storage conditions for the Vitamin A Cassava, such as temperature and humidity requirements. This information ensures that the Vitamin A Cassava are stored under optimal conditions to maintain quality.
- **Quality Grade:** Indicate the quality grade of the Vitamin A Cassava, if applicable. Grades are typically based on factors like size, shape, appearance, and absence of defects.
- **Organic or Conventional:** If the Vitamin A Cassava are organic, label them accordingly. This is important for customers seeking organic produce and for compliance with organic certification standards.
- **Weight or Quantity:** Specify the weight or quantity of Vitamin A Cassava in the batch. This helps in inventory management and facilitates accurate record-keeping.
- **Barcodes or QR Codes:** Consider using barcodes or QR codes on the labels for automated tracking and inventory systems. This enables quick and efficient identification of batches during storage.
- **Regulatory Compliance:** Ensure compliance with labeling regulations and requirements specific to Vitamin A Cassava in your region or market. Adhere to any mandatory labeling information prescribed by regulatory authorities.

MODULE 2: STORAGE CONDITIONS

3.4. STACKING OF RAW MATERIALS:

- **Ventilated Containers:** Use crates, baskets, or bins that allow air to circulate around the Vitamin A Cassava. Avoid sealed plastic bags or airtight containers, as they can trap moisture and promote rot.

IMAGE OF SAMPLE VENTILATED CONTAINER HERE

- **Single Layer:** Store Vitamin A Cassava in a single layer rather than stacking them. This helps to prevent bruising and allows better air circulation.

IMAGE OF SINGLE LAYER STACKING HERE

- **Cool and Dark Environment:** Vitamin A Cassava should be stored in a cool (around 55-60°F or 13-15°C) and dark place, such as a root cellar, basement, or pantry. Avoid storing them in the refrigerator, as the cold temperature can negatively affect their flavor and texture.
- **Proper Handling:** Handle Vitamin A Cassava with care to avoid bruising or damaging the skin. Gently place them in storage containers and avoid dropping or piling them on top of each other.

3.5. STORAGE PACKAGING AND CONTAINERS

- Several traditional methods can be used for the storage of orange flesh OFSP.
- **Use of ventilated storage structures:** Structures such as mud bricks, woods, bamboo. Ventilation openings or gaps are included to allow for air circulation and preventing moist build up to reduce risk of rot.
- **Hanging baskets:** OFSP can be stored in hanging baskets made of woven fiber of mesh materials. This method keeps the potatoes off the ground and allows for airflow around them. This helps to minimize moist buildup and avoid rot. These baskets are to be hanged in a cool, dry area.
- **Use of wooden crates or bins:** The wooden plates bins are designed with slates or with gaps to allow flow of air. the crates can be kept in well ventilated room to avoid buildup of humidity and allow airflow.
- **Storing in mesh bags:** Mesh bags of breathable materials can be used for storage. The storage bags can be hanged or kept in a well-ventilated room.

COLLAGE OF IMAGES OF THE FOUR METHODS LISTED ABOVE HERE

3.6. STORAGE ARRANGEMENTS AND WALKWAYS

- Single Layer: Arrange the sweet biofortified vitamin A cassavas in a single layer rather than stacking them on top of each other. This helps prevent bruising and allows for better airflow around each biofortified vitamin A cassavas.
- Ventilation: Ensure proper ventilation by using containers or storage options that allow air to circulate around the sweet biofortified vitamin A cassavas. This helps maintain optimal humidity levels and prevents the buildup of moisture that can lead to rot.
- Cool and Dark Environment: Store the sweet biofortified vitamin A cassavas in a cool and dark place, such as a root cellar, basement, or pantry. Aim for a temperature range of around 55-60°F (13-15°C) to prevent sprouting and maintain quality.
- Handle with Care: Handle sweet biofortified vitamin A cassavas gently to prevent damage to their skin. Rough handling or dropping them can lead to bruising, which increases the risk of spoilage.

- Regular Inspection: Periodically check stored sweet biofortified vitamin A cassavas for signs of rot or decay. Remove any damaged or spoiled biofortified vitamin A cassavas promptly to prevent the spread of spoilage to other tubers.

3.7. USE OF PALLETS

Pallets are a practical option for storing orange-fleshed sweet biofortified vitamin A cassava. They provide organization, maximize space, allow airflow, and facilitate handling and transportation. Choose sturdy pallets, place a protective layer, stack sweet biofortified vitamin A cassavas in a single layer, and store them in a cool, dark area with proper conditions. Regularly monitor the biofortified vitamin A cassavas for freshness.

IMAGE OF PALLETS AND PALLET USE FOR VIT A CASSAVA STORAGE HERE

MODULE 3: ENVIRONMENT CONDITIONS

3.8. PEST CONTROL

3.8.1. IN-PROCESS OR PREVENTIVE PEST CONTROL

In-process or preventive pest control refers to the implementation of measures and practices to prevent, detect, and control pests within the production process of a food factory. This helps ensure that pests do not contaminate or compromise the safety and quality of the food products. Key considerations for in-process pest control:

1. **Pest Prevention:** Establish a robust pest prevention program that includes regular inspections, monitoring, and identification of potential pest entry points within the production area. Implement structural and physical measures to prevent pests from entering the facility, such as installing door sweeps, sealing cracks and gaps, and using screens on windows and vents. Maintain a clean and clutter-free environment to minimize potential harborage areas for pests
2. **Integrated Pest Management (IPM):** Implement an IPM approach that combines multiple pest control methods, including preventive measures, monitoring, sanitation, physical controls, and targeted pesticide application when necessary. Set action thresholds to determine when pest control measures should be taken based on the presence or level of pest activity. Regularly monitor and document pest activity through visual inspections, trapping, and use of pest monitoring devices.
3. **Sanitation Practices:** Follow strict sanitation practices throughout the facility, including regular cleaning of equipment, floors, walls, and other surfaces to eliminate potential food sources for pests. Properly dispose of waste and garbage in sealed containers to minimize pest attraction. Ensure proper drainage and minimize standing water to prevent pest breeding sites.
4. **Employee Education and Training:** Train employees on the importance of pest control, their role in preventing pest infestations, and the proper procedures for reporting and addressing pest-related issues. Foster a culture of awareness and accountability among employees to promptly report any signs of pests or conditions that may contribute to pest problems.
5. **Monitoring and Documentation:** Implement a systematic monitoring program to detect signs of pest activity, such as droppings, gnaw marks, or sightings, and record findings. Maintain accurate records of pest control activities, including inspections, treatments, and corrective actions taken, as well as any recommendations provided by pest control professionals.
6. **Collaboration with pest control professionals:** Establish a partnership with a licensed pest control company to conduct regular inspections, provide expert advice, and implement targeted pest control measures. Collaborate with pest control professionals to develop a customized pest control plan based on the specific needs and challenges of the food factory.
7. **Regular Audits and Reviews:** Conduct regular internal audits and reviews of the in-process pest control program to assess its effectiveness, identify areas for improvement, and ensure compliance with regulations and industry standards.:

3.8.2. CORRECTIVE PEST CONTROL

Refers to the actions taken to address and rectify issues related to pest infestations or other non-compliance with pest control measures. While preventive pest control focuses on proactive measures to prevent pests, corrective control comes into play when pests have already been detected or when there are breaches in the preventive measures. Here are some key aspects of corrective control:

1. **Pest Identification and Assessment:** Properly identify the pests present in the facility to determine the extent of the infestation and potential risks associated with them. Assess the areas affected by the infestation, including identifying the source of the problem and evaluating the extent of damage or contamination caused.
2. **Immediate Action:** Take prompt action to eliminate the pests and address the specific issues identified during the assessment. Depending on the severity of the infestation, this may involve measures such as targeted pesticide application, trapping, removal of pest habitats, or physical removal of pests.
3. **Root Cause Analysis:** Conduct a thorough investigation to identify the underlying causes that allowed the pest infestation to occur. Determine if there were any gaps or failures in the preventive pest control measures, such as structural deficiencies, sanitation lapses, or employee practices.
4. **Corrective Measures and Remediation:** Develop and implement corrective actions to address the identified root causes and prevent future pest infestations. This may include repairing structural issues, improving sanitation protocols, enhancing employee training, or modifying processes to eliminate pest-attracting conditions.

3.8.3. FUMIGATION

Fumigation is a method of pest control that involves the use of gaseous pesticides, known as fumigants, to eliminate pests in an enclosed space or within certain commodities. It is typically used to control pests such as stored product insects, rodents, and insects that infest structures or equipment.

Fumigation requires specialized knowledge and equipment, and it is usually performed by licensed professionals. The process involves sealing the area or commodity to be fumigated, introducing the fumigant gas, and allowing sufficient time for the gas to penetrate and kill the pests.

Fumigated areas or commodities need to be properly aerated to remove the fumigant gas residue and ensure it is safe for workers and consumers.

3.8.4. USE OF PESTICIDES

Pesticides are chemical substances designed to control or eliminate pests, including insects, rodents, and other pests that can pose a threat to food safety and product quality. Pesticide control in food factories involves the strategic application of approved pesticides to target pests and minimize their impact on the facility and products.

It is crucial to use pesticides approved for use in food processing facilities and comply with regulatory requirements regarding their application and usage. Pesticides should only be applied by trained and licensed personnel following label instructions, safety precautions, and appropriate application methods.

Integrated Pest Management (IPM) principles should be considered to minimize reliance on pesticides and employ a comprehensive approach that includes prevention, monitoring, and non-chemical control methods.

IMAGE COLLAGE OF PESTICIDE WARNING AND USE FOR FOOD INDUSTRY HERE

3.9. TEMPERATURE AND HUMIDITY CONTROL

Proper temperature and humidity control are essential for maintaining the quality and prolonging the shelf life of Vitamin A Cassava. Here are the recommended temperature and humidity conditions for storage:

- **Temperature:** Store Vitamin A Cassava in a cool environment with temperatures ranging from 55-60°F (13-15°C). This temperature range helps slow down the rate of sprouting and maintains the quality of the potatoes.
- **Humidity:** The ideal humidity level for Vitamin A Cassava is around 85-90%. This range helps prevent the potatoes from drying out and shriveling. However, it's important to avoid excessive humidity, as it can promote the growth of mold or rot.

3.10. LIGHT AND VENTILATION CONTROL

- **Light:** Biofortified vitamin A cassavas should be stored in a dark or low-light environment. Exposure to light can cause greening of the biofortified vitamin A cassavas, which affects their flavor and quality. Keep them away from direct sunlight or bright artificial light sources.
- **Ventilation:** Proper ventilation is crucial to prevent the buildup of moisture and maintain optimal airflow around the sweet biofortified vitamin A cassavas. Sufficient airflow helps reduce the risk of rot and keeps the biofortified vitamin A cassavas fresh.

3.11. VITAMIN A CASSAVA RAW MATERIAL STORAGE ENVIRONMENT

Some key factors to consider before and during storage of biofortified vitamin A cassava include:

- **Sorting:** After harvesting, undamaged and healthy cassava should be kept separate for storage. rot cassava should not be stored with undamaged and healthy ones. Defective cassava must be removed to be processed immediately (defective will include broken, exposed tissues etc.) Sorting of potatoes must be done before receiving into storage. Quantity of potatoes to be stored should be recorded
- **Cleaning:** After harvest, sand on surface of cassava can be dusted off before proceeding to store. Only firm and plump cassava should be considered for storage.
- **Temperature and humidity:** It is essential to important to find a cool storage location. Maintain a relative humidity level of 85% to 90% to prevent moist loss from the potatoes
- **Ventilation and air circulation:** This is crucial to prevent moist build up and reduce risk of mold or rot.
- **Pest control:** Take measures to prevent pest infestation during storage. Inspect for signs of pest or damage before storage.
- **Inspection:** Regularly inspect stored cassava for any sign of rot, and if there is rot, separate from the other stored cassava.

3.12. AREA ACCESS CONTROL

Access control for process in food factories refers to the implementation of security measures and protocols to control and restrict access to specific areas within the factory where food processing takes place. This is done to ensure food safety, prevent contamination, maintain quality standards, and comply with regulatory requirements. Some common practices for access control in food factories:

1. **Physical Barriers:** Install physical barriers such as fences, walls, and locked gates to limit access to authorized personnel only. Clearly mark restricted areas and use signage to indicate the need for authorization.

2. **Access Cards and Key Systems:** Implement an access control system that requires employees to use access cards or keys to enter restricted areas. These systems can track and record entry and exit times, providing a log of who accessed the area.
3. **Biometric Authentication:** Use biometric technologies like fingerprint or iris scanners to verify the identity of individuals before granting access to sensitive areas. Biometrics offer a higher level of security and prevent unauthorized use of access cards or keys.
4. **Video Surveillance:** Install surveillance cameras at entry points and throughout the facility to monitor activity and deter unauthorized access. Video footage can also be used for investigations and audits.
5. **Visitor Management:** Implement a visitor management system that requires visitors to sign in, wear identification badges, and be accompanied by authorized personnel while in restricted areas. Escorting visitors helps ensure they do not enter unauthorized areas.
6. **Employee Training:** Provide comprehensive training to employees on access control procedures, including the importance of not sharing access credentials, reporting suspicious activities, and following proper protocols when accessing restricted areas.
7. **Access Control Policies:** Develop and enforce access control policies that clearly outline the rules and procedures for accessing restricted areas. Regularly review and update these policies to align with changing requirements and best practices.
8. **Audit Trails and Monitoring:** Implement systems to track and monitor access events, including failed access attempts. Regularly review audit trails to identify any anomalies or potential security breaches.
9. **Regular Maintenance and Inspections:** Conduct regular maintenance and inspections of access control systems to ensure they are functioning properly. Address any issues promptly to maintain the integrity of the access control measures.
10. **Compliance with Regulations:** Ensure that access control measures align with relevant food safety regulations, such as those set by local health departments, food safety agencies, and industry standards like Hazard Analysis and Critical Control Points (HACCP).

4. A: MANUFACTURING PROCESSES: MACHINERY AND MAINTENANCE

MODULE 1: MACHINERY REQUIREMENTS

4.1. TYPES OF MACHINES REQUIRED

4.1.1. BIOFORTIFIED VITAMIN A CASSAVA CHIPS PROCESSING MACHINE

This is the simplest biofortified vitamin A cassava processing machine. If you want to process biofortified vitamin A cassava chips for animal feed, a biofortified vitamin A cassava chipping machine can get the work done.

The equipment includes:

1. biofortified vitamin A cassava peeler
2. biofortified vitamin A cassava chipping machine, and
3. biofortified vitamin A cassava chips dryer.

Put into consideration that some of these can be done manually or traditionally e.g., washing and peeling.

4.1.2. BIOFORTIFIED VITAMIN A GARRI PROCESSING MACHINE

The complete set of machines for garri processing include:

1. Cassava cleaner
2. Washer, peeler

3. Cassava grater
4. Hydraulic press
5. Garri fryer
6. Vibration sieve and
7. Packaging machine.

Some of these can be done manually or traditionally e.g., washing and peeling.

4.1.3. BIOFORTIFIED VITAMIN A CASSAVA FLOUR PROCESSING MACHINE

The process of biofortified vitamin A cassava flour production is cleaning, washing and peeling, crushing, de-sanding, dewatering, drying, sieving and packaging. Accordingly, the equipped machines are

1. Dry sieve
2. Paddle washer
3. Rasper
4. Desander
5. Plate frame filter press
6. Flash dryer
7. Sieving machine
8. Packaging machine.

Some of these can be done manually or traditionally e.g., washing and peeling.

4.1.4. BIOFORTIFIED VITAMIN A CASSAVA STARCH PROCESSING MACHINE

In a biofortified vitamin A cassava starch processing plant, the whole process is divided into three parts: raw material cleaning section, starch extraction section and drying section. The mainly used equipment includes:

1. Dry sieve
2. Paddle washer
3. Rasper
4. Centrifugal sieve
5. Fine fiber sieve
6. Desander
7. Hydrocyclone system
8. Peeler centrifuge
9. Flash dryer.

Some of these can be done manually or traditionally e.g., washing and peeling.

4.2. CONSIDERATIONS TO CHOSE MACHINERY

When selecting machineries for biofortified vitamin A cassava processing, several factors should be considered, including:

- Processing Capacity: Ensure that the machinery can handle the desired volume of biofortified vitamin A cassava processing.
- Efficiency and Productivity: Look for machines that offer high efficiency and productivity to optimize production.
- Quality of Output: Consider the quality of the end product the machinery can deliver.
- Cost and Affordability: Evaluate the machinery's cost, including maintenance and operational expenses, to ensure it aligns with your budget.
- Energy Requirements: Check the energy consumption of the machines to assess their environmental impact and operational costs.
- Maintenance and Technical Support: Consider the availability of spare parts, maintenance requirements, and technical support from the manufacturer.
- Consideration of cleaning-in-place method applicable to the machine.
- Nutrient Preservation: In choosing machineries consider machineries that minimize nutrient losses to preserve the enhanced nutritional value. This may involve using processing techniques that minimize exposure to heat, air, light, or other factors that can degrade or diminish the nutrient content.
- Quality Control: In choosing machineries, consider machineries that can maintain the nutritional value throughout the processing. It is recommended that regular testing and monitoring should be conducted to assess nutrient levels and verify compliance with desired nutritional standards.

4.3. TRAINING AND TEST RUNNING

Operating biofortified vitamin A cassava processing machineries effectively requires appropriate training. Training should cover the following aspects:

- Machinery Operation: Familiarize operators with the functioning of each machine, including startup, shutdown, adjustments, and routine maintenance.
- Safety Procedures: Educate operators on safety protocols to prevent accidents and ensure a safe working environment.
- Maintenance and Troubleshooting: Train operators on basic maintenance tasks, such as cleaning, lubrication, and minor repairs. Also, provide guidance on identifying and resolving common issues that may arise during operation.
- Quality Control: Train operators to monitor and control the quality of the processed biofortified vitamin A cassava products, ensuring adherence to standards.

MODULE 2: CLEANING IN PLACE CONCEPT

4.4. CONCEPT OF CIP

Cleaning-in-Place (CIP) is defined as "*The cleaning of complete items of plant or pipeline circuits without dismantling or opening of the equipment and with little or no manual involvement on the part of the operator. The process involves the jetting or spraying of surfaces or circulation of cleaning solutions through the plant under conditions of increased turbulence and flow velocity*"

The CIP (Cleaning-in-place) procedures for biofortified vitamin A cassava processing machinery can vary depending on the specific equipment involved. It is important to create SOPs for all processing machinery CIPs. Examples of some common CIP steps for biofortified vitamin A cassava processing machinery:

1. Disassembly: If applicable, disassemble the machinery according to the manufacturer's instructions. This allows for easier access to all parts and facilitates thorough cleaning.

2. Dry Cleaning: Begin by removing any dry debris or residue from the machinery. This can be done using brushes, compressed air, or vacuuming equipment to dislodge and remove loose particles.
3. Flushing: Start the CIP process by flushing the machinery with water to remove any remaining debris or loose material. High-pressure water jets or sprayers can be used to ensure thorough cleaning of surfaces, crevices, and hard-to-reach areas.
4. Cleaning Solution Application: Apply a suitable cleaning solution or detergent to the machinery. The choice of cleaning solution may vary based on the type of machinery and the contaminants to be removed. Follow the manufacturer's recommendations for the appropriate concentration and contact time.
5. Circulation and Soaking: Circulate the cleaning solution throughout the machinery using pumps or recirculation systems. This ensures that the solution reaches all surfaces and components. If necessary, allow the cleaning solution to soak for a specified period to enhance the removal of stubborn residues.
6. Mechanical Action: Some machinery may have surfaces that require mechanical action for effective cleaning. This can include using brushes, scrubbers, or spray devices to agitate and dislodge residues from surfaces or parts.
7. Rinse: After the cleaning solution has had sufficient contact time, thoroughly rinse the machinery with clean water. Ensure that all traces of the cleaning solution and loosened contaminants are removed. Multiple rinsing cycles may be necessary to achieve the desired cleanliness.
8. Sanitization: In addition to cleaning, sanitization is important to eliminate or reduce microbial contamination. Apply a suitable sanitizing agent or disinfectant to the machinery, following recommended concentrations and contact times. This step helps ensure hygienic conditions and prevents the growth of harmful microorganisms.
9. Final Rinse: After sanitization, perform a final rinse with clean water to remove any residual sanitizing agent or disinfectant.
10. Reassembly and Inspection: Once the machinery is thoroughly cleaned and rinsed, reassemble the disassembled parts as per the manufacturer's instructions. Inspect the equipment for any signs of damage, wear, or malfunctions that may require attention or repairs before reuse.

Sample Cleaning Record

Date	Machine	Type of Cleaning	Personnel Responsible	Remarks

____

Quality Control Officer

4.5. ACCEPTABLE CLEANING AGENTS

When it comes to selecting cleaning agents for vitamin A cassava processing machines, it is essential to choose products that are safe, effective, and approved for use in food processing environments. Some commonly used and acceptable cleaning agents for vitamin A cassava processing machines are:

1. Alkaline cleaners: Alkaline cleaners are effective in removing oil, grease, and organic residues. They are commonly used for cleaning vitamin A cassava milling machines, silos, and conveyors. Sodium hydroxide (caustic soda) and potassium hydroxide are examples of alkaline cleaning agents.
2. Acidic cleaners: Acidic cleaners are useful for removing mineral deposits, scale, and inorganic residues. They are often employed for descaling and cleaning equipment such as boilers, heat exchangers, and pipes. Common acidic cleaning agents include citric acid, phosphoric acid, and hydrochloric acid. It is important to handle acidic cleaners with care and follow safety guidelines to prevent corrosion and ensure worker safety.
3. Enzymatic cleaners: Enzymatic cleaners are effective in breaking down protein-based stains and residues. They can be beneficial for removing organic matter from vitamin A cassava processing equipment. Enzymes like proteases or amylases are commonly used in enzymatic cleaning agents.
4. Chlorine-based sanitizers: Chlorine-based sanitizers are used to disinfect and sanitize vitamin A cassava processing equipment. They are effective against a wide range of microorganisms. Sodium hypochlorite (household bleach) or chlorine dioxide are commonly used chlorine-based sanitizers. It is important to use them according to recommended concentrations and rinse thoroughly afterward to remove any residual chlorine.
5. Quaternary ammonium compounds (Quats): Quats are commonly used as sanitizers in the food industry, including vitamin A cassava processing. They are effective against a variety of bacteria, viruses, and fungi. Quats have good stability and are often used for equipment surfaces, floors, and walls.
6. Peroxyacetic acid (PAA): PAA is a potent sanitizer and disinfectant that is effective against a broad spectrum of microorganisms, including bacteria, yeast, molds, and viruses. It is often used in food processing facilities, including vitamin A cassava processing plants, to ensure microbial control.

4.6. SCHEDULED CLEANING BEYOND CIP

In addition to regular Clean-In-Place (CIP) procedures, scheduled cleaning beyond CIP is necessary to maintain optimal hygiene and prevent the buildup of contaminants in orange-fleshed sweet biofortified vitamin A cassava processing machines. Here are specific examples of scheduled cleaning tasks that can be performed. By implementing these specific cleaning tasks, you can ensure the cleanliness and hygiene of orange-fleshed sweet biofortified vitamin A cassava processing machines, reducing the risk of contamination, and maintaining product quality.:

1. Disassembly and manual cleaning: Some parts of orange-fleshed sweet biofortified vitamin A cassava processing machines, such as screens, rollers, and other removable components, may require disassembly for thorough cleaning. Soak, scrub, or use specialized brushes to remove stubborn residues or deposits. Ensure proper reassembly after cleaning.
2. Inspection and cleaning of filters: Filters in orange-fleshed sweet biofortified vitamin A cassava processing equipment remove impurities and foreign particles. Regularly inspect and clean the filters to prevent clogging and ensure uninterrupted operation. Follow the manufacturer's guidelines for proper filter cleaning and replacement.

3. Conveyor belt cleaning: Orange-fleshed sweet biofortified vitamin A cassava processing often involves the use of conveyor belts that can accumulate dust, debris, and product residues. Scheduled cleaning of conveyor belts using appropriate methods such as brushing or vacuuming is essential to maintain cleanliness and prevent cross-contamination.

4. Cleaning of hoppers and storage bins: Regularly clean hoppers and storage bins used in orange-fleshed sweet biofortified vitamin A cassava processing to remove remaining biofortified vitamin A cassava pieces, dust, or debris. Use appropriate tools and equipment for thorough cleaning.
5. Equipment exterior and surrounding area cleaning: In addition to internal cleaning, it is important to clean the exterior surfaces of orange-fleshed sweet biofortified vitamin A cassava processing machines. Wipe down surfaces with suitable cleaning agents to remove dirt, dust, and contaminants. Keep the surrounding area clean and free from debris to maintain a hygienic environment.
6. Regular sanitation of the processing area: Maintaining a clean processing area is crucial for orange-fleshed sweet biofortified vitamin A cassava processing. Regularly sanitize floors, walls, and other surfaces to prevent the growth and spread of bacteria, molds, and other microorganisms. Use appropriate sanitizing agents and follow proper cleaning protocols.

4.7. CONFIRMING HYGIENE STATUS OF EQUIPMENT

By following these steps, you can assess the hygiene status of equipment in a vitamin A cassava processing facility and take appropriate measures to maintain cleanliness and compliance. To confirm the hygiene status of equipment in a vitamin A cassava processing facility, follow these steps:

1. Visual inspection: Check the equipment for visible signs of dirt, residue, or damage, paying attention to areas where contaminants may accumulate.
2. Swab testing: Collect samples from selected surfaces for microbiological analysis using sterile swabs. Send the samples to a laboratory to detect the presence of microorganisms.
3. ATP testing: Use an ATP testing device to measure organic material levels on equipment surfaces. High ATP levels indicate inadequate cleaning and hygiene.
4. Allergen testing: If handling allergenic ingredients, test for cross-contamination using rapid test kits or laboratory analysis.
5. Sensory evaluation: Conduct sensory evaluation to detect any abnormal odors or tastes that may indicate contamination. Trained personnel can assess the expected sensory attributes of vitamin A cassava products.
6. Review cleaning records: Ensure proper cleaning procedures, including CIP protocols, scheduled tasks, and cleaning agent usage, are followed and documented consistently.
7. Maintenance inspection: Regularly inspect and maintain equipment to ensure proper functioning and cleanliness. Address any issues promptly.
8. Regulatory compliance: Ensure equipment meets food processing hygiene standards and comply with local health and safety regulations.

MODULE 3: MACHINE PREVENTIVE MAINTENANCE

4.8. PREVENTIVE MAINTENANCE

Preventive Maintenance for biofortified vitamin A cassava Processing Machinery

1. Create a Maintenance Schedule: Develop a maintenance schedule that outlines regular maintenance tasks for each machinery involved in biofortified vitamin A cassava processing. This schedule should include daily weekly, monthly, and annual maintenance activities.
2. Regular Cleaning: Regularly clean the machinery to remove dirt, debris, and residual materials. This helps prevent buildup and potential blockages that can affect the performance and longevity of the machines.
3. Lubrication: Apply appropriate lubricants to the moving parts of the machinery as recommended by the manufacturer. Lubrication reduces friction, minimizes wear and tear, and helps maintain smooth operation.
4. Inspection and Adjustment: Conduct routine inspections of the machinery to identify any loose or worn-out components. Tighten loose bolts and nuts, replace damaged or worn-out parts, and make necessary adjustments to ensure optimal performance.
5. Belt and Chain Maintenance: Inspect belts and chains for signs of wear, stretching, or misalignment. Replace worn-out belts and adjust chain tension as necessary to maintain proper functioning.
6. Calibration and Alignment: Periodically calibrate and align the machinery to ensure accurate and consistent performance. This is particularly important for machines involved in grading, slicing, or cutting biofortified vitamin A cassava.
7. Electrical Connections: Check electrical connections, wires, and cables for any signs of damage or loose connections. Tighten connections and replace damaged components to prevent electrical issues and potential hazards.
8. Safety Devices: Test and verify the functionality of safety devices such as emergency stop buttons, safety guards, and interlocks. Ensure they are in good working condition and promptly address any malfunctions or damage.
9. Documentation and Record-Keeping: Maintain a comprehensive record of maintenance activities, including dates of maintenance tasks performed, parts replaced, and any observations or issues identified. This record helps track maintenance history and facilitates future troubleshooting or planning.
10. Manufacturer's Recommendations: Follow the manufacturer's guidelines and recommendations for preventive maintenance. Consult the machinery's user manual or contact the manufacturer directly for specific maintenance schedules, procedures, and recommendations.
11. Operator Training: Provide training to operators on basic maintenance tasks, such as cleaning, lubrication, and routine inspections. Educate them on identifying early signs of issues or abnormalities and encourage reporting to facilitate timely maintenance interventions.

4.8.1. ADVANTAGES OF PREVENTIVE MAINTENANCE

Implementing preventive maintenance for biofortified vitamin A cassava machineries offer several advantages:

- Increased Equipment Reliability: Regular maintenance tasks help identify and address potential issues before they turn into major breakdowns. By conducting inspections, lubrication, and adjustments, preventive maintenance minimizes the chances of unexpected failures or downtime during biofortified vitamin A cassava processing operations. This improves the reliability of the machinery.
- Extended Equipment Lifespan: Preventive maintenance helps preserve the condition of biofortified vitamin A cassava machineries over time. By regularly cleaning, lubricating, and replacing worn-out components, the machinery's lifespan can be extended. This reduces the need for frequent replacements, resulting in cost savings for the business.
- Optimal Equipment Performance: Well-maintained biofortified vitamin A cassava machineries operate at their optimal performance levels. Regular inspections and adjustments ensure that the machines are properly aligned, calibrated, and functioning as intended. This leads to efficient processing, higher productivity, and improved quality of the processed biofortified vitamin A cassava products.
- Minimized Downtime: Equipment breakdowns or failures can result in significant production delays and downtime. Preventive maintenance aims to detect and address issues proactively, reducing the likelihood of sudden breakdowns. By scheduling

maintenance tasks during planned downtime, the impact on overall production can be minimized, ensuring continuous operation.

- Cost Savings: Preventive maintenance helps reduce repair and replacement costs. By addressing minor issues before they escalate, the need for costly repairs or component replacements can be avoided. Additionally, extending the lifespan of the machinery through regular maintenance reduces the capital expenditure required for purchasing new equipment.
- Enhanced Safety: Regular maintenance includes inspecting safety features and ensuring their proper functioning. This helps maintain a safe working environment for operators and reduces the risk of accidents or injuries. Well-maintained machines are less likely to have malfunctions that could pose safety hazard.
- Improved Product Quality: Maintaining biofortified vitamin A cassava processing machinery in optimal condition ensures consistent and reliable processing. This leads to improved product quality and reduces the chances of defects or inconsistencies in the processed biofortified vitamin A cassava products. Enhanced product quality can contribute to customer satisfaction and market competitiveness.
- Compliance with Regulations: Depending on the region or industry, there may be regulatory requirements for maintaining and operating machinery. Implementing preventive maintenance helps ensure compliance with these regulations, avoiding potential penalties or legal issues (SON inspects preventive maintenance of Machines).

4.9. PLANNING MAINTENANCE

Planning maintenance for biofortified vitamin A cassava processing machineries involves a systematic approach to ensure timely and effective maintenance activities. Here are the steps involved in planning maintenance:

1. Identify Maintenance Needs: Assess the maintenance requirements of each biofortified vitamin A cassava processing machinery. Consider factors such as the manufacturer's recommendations, historical data on maintenance issues, and the machinery's critical components.
2. Develop a Maintenance Schedule: Create a schedule that outlines the frequency and timing of maintenance tasks for each machinery. Consider the equipment's operational hours, production cycles, and any seasonal fluctuations in demand.
3. Prioritize Maintenance Tasks: Determine the priority of each maintenance task based on its impact on equipment performance, production efficiency, and safety. Focus on critical tasks that directly affect production or have the potential to cause significant disruptions if not addressed promptly. (Ensure you create time to shut down the system if need be to carry out maintenance)
4. Allocate Resources: Determine the necessary resources for maintenance, including personnel, tools, spare parts, and any external support required. Ensure that the resources are available and allocated accordingly to execute the planned maintenance activities.
5. Create a Work Order System: Establish a system for generating work orders that specify the maintenance tasks, responsible personnel, timelines, and any special instructions or precautions. This helps streamline the maintenance process and ensures clear communication among team members.
6. Procure and Organize Spare Parts: Identify the required spare parts for maintenance activities and ensure their availability. Establish a system for procuring, organizing, and tracking spare parts to minimize downtime caused by waiting for replacements.
7. Document Maintenance Procedures: Create detailed maintenance procedures and guidelines for each machinery. Include step-by-step instructions, safety precautions, and recommended tools or techniques. Documenting procedures ensures consistency, facilitates training, and serves as a reference for future maintenance activities.
8. Implement a Maintenance Tracking System: Utilize a maintenance tracking system to record and monitor maintenance activities. This system should include information such as maintenance dates, tasks performed, observations made, and any

follow-up actions required. The tracking system aids in tracking maintenance history, identifying recurring issues, and planning future maintenance tasks.

9. Review and Improve: Regularly review the effectiveness of the maintenance plan and seek feedback from maintenance personnel. Analyze maintenance data and identify areas for improvement, such as optimizing schedules, refining procedures, or investing in additional training or resources.
10. Maintain Communication and Collaboration: Foster communication and collaboration among maintenance personnel, production teams, and management. Encourage reporting of equipment issues or abnormalities, feedback on maintenance processes, and suggestions for improvement.

4.10. MAINTAINING SPARE-REORDER LEVEL

Setting a maintenance spare reorder level ensures that an adequate supply of spare parts is maintained to support ongoing maintenance activities for biofortified vitamin A cassava biofortified vitamin A cassava processing machineries. The reorder level is the quantity of spare parts that triggers the initiation of the procurement process to replenish the stock. To determine the maintenance spare reorder level for orange-fleshed sweet biofortified vitamin A cassavas, follow these steps:

- Analyze historical usage and determine the average consumption rate for each spare part.
- Consider the criticality of equipment and associated spare parts for orange-fleshed sweet biofortified vitamin A cassavas.
- Determine the lead time required for procuring spare parts.
- Calculate the safety stock to account for unexpected variations in demand or lead time.
- Calculate the reorder point by adding the average consumption rate multiplied by the lead time to the safety stock.
- Set the reorder level slightly above the reorder point, considering ordering frequency, procurement lead time, and storage capacity.
- Monitor inventory levels to ensure they don't fall below the reorder level.
- Periodically review and adjust the reorder level based on changes in equipment performance, maintenance requirements, or other factors.

4.11. IN-HOUSE AND OUTSOURCING CRITERIA

4.11.1. IN-HOUSE MAINTENANCE CRITERIA

1. Expertise and Resources: Assess whether your organization has the necessary expertise, skills, and resources to handle maintenance tasks effectively. This includes having trained personnel, access to tools and equipment, and the capability to address a wide range of maintenance needs.
2. Equipment Knowledge: Consider the level of familiarity your in-house team has with the specific biofortified vitamin A cassava processing machineries. If your team is highly knowledgeable about the equipment, they may be better equipped to identify issues and perform maintenance tasks efficiently.
3. Cost: Evaluate the cost implications of maintaining an in-house maintenance team. This includes salaries, benefits, training, tools, and equipment. Compare these costs to potential savings achieved through timely maintenance, reduced downtime, and optimized equipment performance.
4. Control and Flexibility: In-house maintenance offers greater control over scheduling, prioritization, and decision-making. It allows you to tailor maintenance activities to your specific needs and adjust plans as necessary. Having an in-house team also provides more immediate response to maintenance emergencies.

4.11.2. OUTSOURCED MAINTENANCE CRITERIA

1. Specialized Expertise: Determine if there are specific maintenance tasks or specialized skills required for your biofortified vitamin A cassava processing machineries that your in-house team may lack. Outsourcing to specialized maintenance service providers can ensure that experts with in-depth knowledge and experience in the field handle critical maintenance tasks.
2. Cost Efficiency: Consider the potential cost savings of outsourcing maintenance. Outsourced maintenance can provide access to shared resources, bulk purchasing power, and economies of scale, potentially reducing overall maintenance costs compared to maintaining an in-house team.
3. Focus on Core Competencies: Outsourcing maintenance allows your organization to focus on its core competencies and primary business activities. By entrusting maintenance tasks to external experts, you can allocate more resources and attention to your primary operations and strategic goals.
4. Scalability and Flexibility: Outsourcing offers the advantage of scalability, allowing you to adjust the level of maintenance support based on your needs. You can engage maintenance service providers for specific tasks, periodic support, or comprehensive maintenance contracts, providing flexibility to adapt to changing demands.
5. Access to Advanced Technology: Maintenance service providers often invest in advanced technologies, diagnostic tools, and software systems that may be cost-prohibitive for an individual organization. By outsourcing, you can leverage their access to such technologies, which can enhance the effectiveness and efficiency of maintenance activities.
6. Reduced Administrative Burden: Outsourcing maintenance can relieve your organization of administrative tasks associated with managing an in-house maintenance team. This includes recruitment, training, performance management, and compliance with labor regulations.

4.12. HYGIENE DURING MAINTENANCE

Maintaining proper hygiene during maintenance activities is essential to ensure food safety and prevent contamination. Some guidelines for promoting hygiene during maintenance:

1. Personal Hygiene: Ensure that maintenance personnel adhere to good personal hygiene practices. This includes wearing clean and appropriate protective clothing, such as gloves, hairnets, and clean uniforms. Encourage regular handwashing with soap and water before and after maintenance tasks.
2. Cleaning and Sanitization: Before starting maintenance work, clean the surrounding area to remove any dirt, debris, or contaminants. Use appropriate cleaning agents and sanitizers to maintain cleanliness. Pay extra attention to areas that come into direct contact with the biofortified vitamin A cassava processing machinery or food contact surfaces.
3. Separation of Tools and Equipment: Maintain separate sets of tools and equipment for maintenance activities that involve different areas or stages of biofortified vitamin A cassava processing. This helps prevent cross-contamination between different processing zones.
4. Containment of Contaminants: Use appropriate containment measures to prevent the spread of contaminants during maintenance. For example, use plastic sheeting or barriers to isolate the maintenance area and prevent dust, debris, or other particles from entering the processing zone.
5. Equipment Disassembly and Cleaning: If maintenance tasks require disassembly of equipment, follow proper procedures for dismantling and cleaning. Thoroughly clean and sanitize all parts that come into direct contact with biofortified vitamin A cassava or food products. Ensure that all cleaning agents used are food-grade and approved for use in food processing environments.

6. Allergen Control: If your biofortified vitamin A cassava processing facility handles allergenic substances, take extra precautions during maintenance to prevent cross-contact. Clearly identify and label equipment or areas that may contain allergens and ensure that maintenance personnel are aware of the allergenic risks and necessary preventive measures.
7. Waste Management: Properly dispose of any waste generated during maintenance, including used cleaning materials, lubricants, or damaged parts. Use designated waste disposal containers and follow the appropriate waste management procedures to prevent contamination and maintain a clean environment.
8. Post-Maintenance Cleaning: After completing maintenance tasks, conduct a thorough cleaning of the work area and the equipment involved. Remove any cleaning agents, residues, or debris to restore the cleanliness of the processing environment.
9. Training and Awareness: Provide training to maintenance personnel on the importance of hygiene during maintenance activities. Ensure they understand the potential risks of contamination and the specific procedures and precautions to follow to maintain hygiene standards. Encourage open communication and reporting of any hygiene concerns or incidents.
10. Documentation and Auditing: Maintain records of maintenance activities, including cleaning procedures, cleaning agents used, and any deviations or incidents that occur during maintenance. Regularly conduct internal audits to assess compliance with hygiene practices and identify areas for improvement.

4.13. MACHINE REPLACEMENT AND UPGRADE

Machine replacement and upgrades are important considerations in maintaining efficient and productive biofortified vitamin A cassava processing operations. Here are some factors to consider when deciding whether to replace or upgrade machinery:

1. Age and Condition: Evaluate the age and overall condition of the machinery. Older machines may be more prone to breakdowns, inefficiencies, and increased maintenance needs. If the machinery is nearing the end of its expected lifespan or requires frequent repairs, it may be a sign that replacement or upgrade is necessary.
2. Technological Advances: Assess the technological advancements in the industry and determine if newer machines offer improved capabilities, efficiency, or safety features. Upgrading to more advanced machinery can result in higher productivity, better quality output, and reduced operating costs.
3. Capacity and Scalability: Consider the capacity of the existing machinery and whether it aligns with current and future production needs. If the biofortified vitamin A cassava processing demand is increasing or if the machinery is operating at full capacity, upgrading to larger or more efficient equipment may be beneficial.
4. Cost Analysis: Conduct a thorough cost analysis that compares the expenses associated with replacing or upgrading machinery against the potential benefits. Consider the initial investment, installation costs, ongoing maintenance and operating expenses, and the expected return on investment (ROI) in terms of increased productivity, reduced downtime, or improved product quality.
5. Regulatory Compliance: Stay updated on any changes in regulatory requirements related to biofortified vitamin A cassava processing machinery. If new regulations are implemented that impact the safety, efficiency, or environmental aspects of the machinery, it may be necessary to replace or upgrade the equipment to ensure compliance.
6. Energy Efficiency: Evaluate the energy consumption of the existing machinery. Upgrading to more energy-efficient equipment can result in long-term cost savings through reduced energy expenses and may align with sustainability goals.
7. Vendor Reputation and Support: Research and evaluate the reputation, reliability, and support services of potential equipment vendors or manufacturers. Consider factors such as warranty coverage, maintenance agreements, spare parts availability, and customer reviews. Choose vendors that have a proven track record of delivering quality equipment and reliable after-sales support.

8. Training and Operator Familiarity: Assess the training requirements for operators and maintenance personnel when considering machine replacement or upgrades. Determine the level of training needed to operate and maintain the new equipment effectively. Consider whether the learning curve and training costs can be managed within the transition period.
9. Maintenance and Availability of Spare Parts: Consider the availability of spare parts and the maintenance requirements of the existing machinery. If spare parts are difficult to source or if maintenance is becoming a challenge, it may be more practical to replace the machinery with newer models that have better support and accessibility to spare parts.

4.14. POST MAINTENANCE CHECK CRITERIA

After completing maintenance tasks on biofortified vitamin A cassava processing machinery, it is important to perform post-maintenance checks to ensure that the equipment is functioning properly and ready for operation. Here are some criteria to consider during post-maintenance checks:

1. Visual Inspection: Conduct a visual inspection of the machinery to look for any signs of damage, loose, or missing parts, leaks, or abnormalities. Check for proper alignment of belts, chains, and other moving components.
2. Functionality Testing: Test the machinery to ensure that it is functioning correctly. Run the equipment through its various operational modes or cycles and observe its performance. Check if all controls, sensors, and safety devices are working as intended.
3. Lubrication: Verify that all lubrication points have been properly lubricated as per manufacturer recommendations. Ensure that lubricants are at the appropriate levels and that there are no leaks or excessive grease accumulation.
4. Electrical Systems: Inspect electrical connections, cables, and wiring for any signs of damage, wear, or loose connections. Verify that all electrical components, such as motors, switches, and control panels, are operating correctly.
5. Calibration and Adjustment: If applicable, check and calibrate any sensors, meters, gauges, or measuring devices that were adjusted or replaced during maintenance. Ensure that they are accurate and provide reliable readings.
6. Safety Checks: Test the safety features of the machinery, such as emergency stop buttons, interlocks, and guarding systems, to ensure they are functional and providing adequate protection.
7. Performance Evaluation: Assess the performance of the machinery by comparing it to pre-maintenance benchmarks or expected performance standards. Check if any issues identified during maintenance have been effectively resolved and if the equipment is meeting the required specifications.
8. Documentation and Record keeping: Maintain accurate records of the maintenance activities performed, including details of repairs, replacements, adjustments, and any issues encountered. Update maintenance logs or management systems accordingly.
9. Operator Training: If significant changes were made to the machinery or operating procedures during maintenance, ensure that operators receive appropriate training and instruction on the updated equipment or processes.
10. Post-Maintenance Testing: Depending on the nature of the maintenance performed, conduct additional tests or inspections specific to the equipment to ensure it meets operational requirements and quality standards.

4.15. FOOD GRADE MAINTENANCE MATERIAL

When conducting maintenance on biofortified vitamin A cassava processing machinery in a food processing environment, it is crucial to use food-grade maintenance materials to prevent contamination and ensure food safety. Here are some common food-grade maintenance materials:

1. Lubricants: Choose food-grade lubricants specifically designed for use in food processing equipment. These lubricants are formulated to be non-toxic, odorless, and safe for incidental contact with food. They should meet regulatory standards.

2. Cleaning Agents: Select food-grade cleaning agents suitable for removing grease, dirt, and residues from equipment surfaces. These cleaning agents should be non-toxic, non-corrosive, and approved for use in food processing environments. Follow the manufacturer's instructions for proper dilution and application.
3. Sealants and Adhesives: Use food-grade sealants and adhesives when repairing or replacing gaskets, seals, or joints in biofortified vitamin A cassava processing machinery. Ensure that these materials are suitable for food contact and compliant with relevant food safety regulations.
4. Coatings and Paints: If equipment surfaces require coating or painting, choose food-grade coatings that are resistant to moisture, corrosion, and wear. These coatings should be free of harmful substances and provide a smooth, easy-to-clean surface.
5. Welding Materials: If welding repairs or modifications are necessary, use food-grade welding materials that do not introduce contaminants into the food processing environment. Welding materials should be specifically labeled as suitable for food contact applications.
6. Filters and Strainers: When replacing filters or strainers in processing equipment, select food-grade materials that are safe for use with food products. Ensure that the filters and strainers meet hygiene and filtration requirements.
7. Spare Parts: When sourcing spare parts for biofortified vitamin A cassava processing machinery, choose parts made from food-grade materials, such as stainless steel or food-safe plastics. These parts should be durable, corrosion-resistant, and compliant with food safety regulations.
8. Personal Protective Equipment (PPE): Provide maintenance personnel with food-grade PPE, such as gloves, aprons, and safety glasses, to prevent contamination of equipment surfaces during maintenance activities. Ensure that the PPE materials do not pose a risk to food safety.
9. Storage Containers: Store maintenance materials, such as lubricants, cleaning agents, and spare parts, in food-grade containers or designated storage areas that are clean and free from contaminants. Use properly labeled containers to avoid confusion and cross-contamination.
10. Supplier Certifications: Work with reputable suppliers who can provide documentation and certifications to confirm that the maintenance materials meet food-grade standards. Look for certifications such as NSF International certification or other recognized food safety certifications. It is essential to follow the manufacturer's recommendations and guidelines when using food-grade maintenance materials. Regularly check and verify that the materials used are suitable for food contact and comply with relevant food safety regulations in your specific region.

B - MANUFACTURING PROCESSES: ENVIRONMENT AND CONTROL

MODULE 4: PROCESS ENVIRONMENT

4.16. PROCESS ENVIRONMENT REQUIRMENTS

Creating an appropriate process environment in a processing facility is crucial for food safety and product quality. Consider the following key requirement:

1. **Temperature Control**
   i. Maintain appropriate temperature ranges for different processing stages.
   ii. Prevent microbial growth and ensure food safety.
   iii. Adhere to industry guidelines and regulatory requirements.
2. **Humidity Management**
   i. Control humidity levels to prevent mold growth, caking, and degradation of vitamin A cassava quality.

ii. Implement ventilation, dehumidification, or moisture barriers.
iii. Maintain optimal humidity throughout the processing area.

3. **Air Quality**
   i. Ensure clean and filtered air to minimize contamination risks.
   ii. Use effective air filtration systems to remove airborne particles and allergens.
   iii. Regularly inspect and clean ventilation systems for proper airflow.
4. **Ventilation**
   i. Establish proper ventilation to control odors and maintain a comfortable working environment.
   ii. Dissipate excess heat, moisture, fumes, and airborne contaminants.
   iii. Promote a healthier and safer process environment.
5. **Lighting**
   i. Provide adequate lighting levels for clear visibility and safe operations.
   ii. Choose appropriate lighting fixtures designed for food processing areas.
   iii. Ensure lighting is resistant to breakage and easily cleanable.
6. **Cleanliness and Sanitation**
   i. Implement strict cleanliness and sanitation practices throughout the facility.
   ii. Regularly clean and sanitize processing equipment, surfaces, floors, and walls.
   iii. Follow industry-standard cleaning procedures and use approved cleaning agents for food contact surfaces.
7. **Pest Control:**
   i. Implement effective pest control measures to prevent infestations and product contamination.
   ii. Conduct regular inspections, practice proper waste management, and use approved pesticides if necessary.
   iii. Maintain a comprehensive pest management program.
8. **Waste Management**
   i. Establish proper waste management practices for timely removal and disposal.
   ii. Separate and dispose of waste in designated containers to prevent cross-contamination.
   iii. Maintain a clean and organized process environment.
9. Facility Design and Layout:
   i. Optimize facility design and layout for efficient workflow and food safety.
   ii. Ensure adequate space for equipment, proper segregation of processing areas, and clear separation of raw and finished product zones.
   iii. Include designated handwashing and sanitation stations.
10. **Compliance with Regulatory Standards:**
    i. Adhere to all applicable regulatory standards, guidelines, and certifications related to food processing.
    ii. Stay updated with industry best practices and regulatory changes for ongoing compliance.

Regularly review and update procedures to meet food safety requirements.

4.17. STANDARD OPERATING PROCEDURES

Standard Operating Procedures (SOPs) are detailed, step-by-step instructions that outline the specific processes and activities required to be followed in a food factory. SOPs provide consistency, ensure compliance with regulations, promote safety, and maintain quality standards.

It is crucial to customize SOPs to the specific processes, equipment, and requirements of each food factory. SOPs should be regularly reviewed, updated, and communicated to all relevant personnel. Additionally, training programs should be conducted to ensure employees understand and follow the SOPs consistently. Some examples of processes that require SOPs include:

- Raw material reception
- Different Quality assurance processes -production access protocol
- Machine maintenance
- Cleaning Procedures
- Pest control procedures
- Machine Operation procedures, etc.

4.18. MANUFACTURING AND PRODUCTION AREA ACCESS CONTROL

Just the same way access control is required in raw material storage, as earlier discussed, access control for process in vitamin A cassava processing factories refers to the implementation of security measures and protocols to control and restrict access to specific areas within the factory where food processing takes place. This is done to ensure food safety, prevent contamination, maintain quality standards, and comply with regulatory requirements. Some common practices for access control in food factories:

1. Physical Barriers: Install physical barriers such as fences, walls, and locked gates to limit access to authorized personnel only. Clearly mark restricted areas and use signage to indicate the need for authorization.
2. Access Cards and Key Systems: Implement an access control system that requires employees to use access cards or keys to enter restricted areas. These systems can track and record entry and exit times, providing a log of who accessed the area.
3. Biometric Authentication: Use biometric technologies like fingerprint or iris scanners to verify the identity of individuals before granting access to sensitive areas. Biometrics offer a higher level of security and prevent unauthorized use of access cards or keys.
4. Video Surveillance: Install surveillance cameras at entry points and throughout the facility to monitor activity and deter unauthorized access. Video footage can also be used for investigations and audits.
5. Visitor Management: Implement a visitor management system that requires visitors to sign in, wear identification badges, and be accompanied by authorized personnel while in restricted areas. Escorting visitors helps ensure they do not enter unauthorized areas.
6. Employee Training: Provide comprehensive training to employees on access control procedures, including the importance of not sharing access credentials, reporting suspicious activities, and following proper protocols when accessing restricted areas.
7. Access Control Policies: Develop and enforce access control policies that clearly outline the rules and procedures for accessing restricted areas. Regularly review and update these policies to align with changing requirements and best practices.
8. Audit Trails and Monitoring: Implement systems to track and monitor access events, including failed access attempts. Regularly review audit trails to identify any anomalies or potential security breaches.

9. Regular Maintenance and Inspections: Conduct regular maintenance and inspections of access control systems to ensure they are functioning properly. Address any issues promptly to maintain the integrity of the access control measures.
10. Compliance with Regulations: Ensure that access control measures align with relevant food safety regulations, such as those set by local health departments, food safety agencies, and industry standards like Hazard Analysis and Critical Control Points (HACCP).

4.19. USE OF PPE WITHIN PRODUCTION AREA

The use of Personal Protective Equipment (PPE) is essential within the production area of cassava products to ensure worker safety and prevent potential hazards. Here are some key considerations for PPE usage:

1. Head Protection:
   - Wear a suitable safety helmet or bump cap to protect against falling objects, head injuries, or impacts in areas where there is a risk of such incidents.
2. Eye and Face Protection:
   - Use safety goggles or face shields to safeguard the eyes and face from chemical splashes, dust, or debris generated during potato processing operations.
   - When handling corrosive substances or operating machinery that produces airborne particles, ensure appropriate eye and face protection is worn.
3. Respiratory Protection:
   - Depending on the processing tasks involved, respiratory protection may be necessary to prevent inhalation of dust, fumes, or harmful gases.
   - Respirators with suitable filtration capabilities, such as N95 masks or respirators, should be used in accordance with occupational health and safety guidelines.
4. Hand Protection:
   - Wear appropriate gloves, such as cut-resistant or heat-resistant gloves, to protect hands from sharp tools, hot surfaces, or chemical contact.
   - Select gloves that are suitable for the specific tasks, provide adequate dexterity, and offer protection against potential hazards.
5. Body Protection:
   - Use suitable clothing that covers the body to protect against chemical spills, splashes, or contact with hot surfaces during potato processing.
   - Consider wearing aprons, coveralls, or smocks made of appropriate materials that provide resistance to chemicals, heat, or other relevant risks.
6. Foot Protection:
   - Use safety shoes or boots with protective toe caps and slip-resistant soles to prevent foot injuries from heavy objects, sharp tools, or slippery surfaces.
   - Select footwear that is appropriate for the specific work environment, such as waterproof or chemical-resistant boots when handling hazardous substances.
7. Hearing Protection:
   - If working in areas with high noise levels, use earplugs or earmuffs to protect against potential hearing damage.
   - Follow recommended noise exposure limits and wear appropriate hearing protection when required.

4.20. MONITORING AND CONTROL POINTS

Monitoring and control points are specific stages or locations within the food production process where critical control measures are applied to ensure food safety, quality, and regulatory compliance. These points are identified based on potential hazards that could occur during processing and are critical to preventing or minimizing those hazards. Examples of monitoring and control points commonly implemented in food factories include:

- Temperature Control Points
- Time Control Points
- pH Control Points
- Acidic Control Points
- Chemical Control Points
- Allergen Control Points
- Metal Detection or X-ray Inspection Points Packaging Control Points Record-Keeping Points
- Supplier Verification Points

4.21. QUALITY ASSURANCE FLASH POINTS

Flashpoints typically refer to critical areas or activities that require special attention to ensure product quality and adherence to quality standards. These flashpoints are often associated with potential risks or vulnerabilities that can impact the safety, integrity, or quality of the final food products. It is important to continuously assess and review these flashpoints to identify areas for improvement and implement appropriate corrective and preventive actions. Examples of quality assurance flashpoints in food factories include:

1. Raw Material Inspection:
    - Thoroughly inspecting incoming raw materials to ensure they meet quality specifications, including visual appearance, sensory attributes, and adherence to regulatory requirements.
    - Sampling and testing for contaminants, such as pesticides, heavy metals, or microbiological hazards.
2. Allergen Control:
    - Preventing cross-contamination and ensuring proper segregation of allergenic ingredients throughout the production process.
    - Implementing effective cleaning procedures and verifying the absence of allergens on shared equipment or surfaces.
3. Processing Parameters:
    - Monitoring and controlling critical processing parameters, such as temperature, time, pressure, pH, or water activity, to ensure they are within specified limits.
    - Verifying that process deviations are addressed promptly, and corrective actions are taken to maintain product quality and safety.
4. Packaging Integrity:
    - Conducting regular checks to ensure the integrity and functionality of packaging materials, seals, and closures to prevent product contamination, spoilage, or tampering.
    - Implementing quality control measures to detect and address packaging defects or issues that could compromise product quality or safety.

5. Hygiene and Sanitation:
    - Enforcing stringent hygiene practices, including handwashing, proper use of personal protective equipment (PPE), and sanitation of production areas and equipment.
    - Conducting routine inspections and swab testing to verify cleanliness and microbial control.
6. Quality Control Testing:
    - Performing regular quality control tests and analysis on in-process samples and finished products, including sensory evaluation, microbiological testing, chemical analysis, or physical measurements.
    - Ensuring compliance with internal quality standards, regulatory requirements, and customer specifications.
7. Traceability and Recall Management:
    - Establishing robust traceability systems to track and identify the origin of raw materials, ingredients, and finished products.
    - Developing effective recall procedures and conducting mock recall exercises to ensure swift and accurate product retrieval in case of quality or safety issues.
8. Documentation and Record-Keeping:
    - Maintaining comprehensive and accurate documentation of quality control activities, including test results, inspection reports, batch records, and deviations.
    - Regularly reviewing records to identify trends, assess product quality, and drive continuous improvement.
9. Training and Competence:
    - Providing ongoing training and education to employees regarding quality standards, food safety practices, and adherence to standard operating procedures.
    - Ensuring employees are knowledgeable and competent in their roles to maintain product quality and safety.

4.22. ROLES AND RESPONSIBILITIES IN PROCESS ENVIRONMENT

In a process environment, such as a food factory, various roles and responsibilities are assigned to different individuals to ensure smooth operations, maintain quality standards, and promote safety. The specific roles and responsibilities may vary depending on the size and complexity of the facility.

The different responsibilities must be well defined and the roles to carry out these responsibilities must be clearly assigned. For example, the operator of machinery must be known, the personnel assigned to a particular function must be known and all these documented.

Roles such as Production Manager or Quality Assurance Manager are to be assigned to qualified professionals in terms of Training, Education and Experience. The roles can be communicated through Job descriptions, appointment letters or through Standard Operating Procedures where roles are defined and assigned to personnel.

4.23. USE OF FORKLIFT AND HEAVY-DUTY MACHINERY

The use of forklifts and heavy-duty machines in food factories can greatly enhance efficiency, productivity, and material handling capabilities. However, their usage in such environments requires careful consideration and adherence to safety guidelines to prevent accidents, product contamination, and damage to equipment or facilities. Some key points regarding the use of forklifts and heavy-duty machines in food factories:

1. Operator Training: Only trained and authorized personnel should operate forklifts and heavy-duty machines. Operators should receive comprehensive training on equipment operation, safety procedures, load handling techniques, and emergency protocols.
2. Safety Equipment: Operators should wear appropriate personal protective equipment (PPE) such as helmets, safety shoes, high-visibility vests, and gloves. Seat belts must be worn when operating vehicles equipped with them.
3. Maintenance and Inspections: Regular maintenance and inspections are essential to ensure the safe operation of forklifts and heavy-duty machines. This includes checking brakes, tires, steering, lights, and other critical components. Any faults or malfunctions should be promptly addressed.
4. Load Capacity and Stability: Operators must be aware of the load capacity of the equipment and ensure that loads are within safe limits. Proper load distribution and securing methods should be followed to maintain stability during transport.
5. Traffic Management: Establish designated traffic routes and ensure clear signage to separate forklift and heavy machine operations from pedestrian areas. Traffic rules, including speed limits and right-of-way, should be enforced to prevent accidents.
6. Hazard Identification: Conduct a thorough assessment of the workspace to identify potential hazards such as low clearance areas, uneven surfaces, slippery floors, or overhead obstructions. Mitigation measures should be implemented to minimize risks.
7. Material Handling Procedures: Implement proper material handling procedures to prevent product contamination. This includes using clean and sanitized forks or attachments, avoiding contact between the equipment and food products, and preventing spills or leaks.
8. Fire Safety: Ensure that forklifts and heavy-duty machines are equipped with fire extinguishers, and operators should be trained on how to use them. Strict no-smoking policies should be enforced in designated areas.
9. Emergency Preparedness: Establish emergency response plans and provide training to operators on how to respond to accidents, spills, or equipment failures. Emergency exits and evacuation routes should be clearly marked.
10. Environmental Considerations: Proper waste disposal procedures should be followed, and any spills or leaks of hazardous materials should be immediately reported and addressed.

MODULE 5: PEST CONTROL IN PROCESS ENVIRONMENT

4.24. PREVENTIVE AND CORRECTIVE CONTROL

Preventive or in-process pest control refers to the implementation of measures and practices to prevent, detect, and control pests within the production process of a food factory. This helps ensure that pests do not contaminate or compromise the safety and quality of the food products. Key considerations for in-process pest control:

1. Pest Prevention:
    - Establish a robust pest prevention program that includes regular inspections, monitoring, and identification of potential pest entry points within the production area.
    - Implement structural and physical measures to prevent pests from entering the facility, such as installing door sweeps, sealing cracks and gaps, and using screens on windows and vents.
    - Maintain a clean and clutter-free environment to minimize potential harborage areas for pests
2. Integrated Pest Management (IPM):
    - Implement an IPM approach that combines multiple pest control methods, including preventive measures, monitoring, sanitation, physical controls, and targeted pesticide application when necessary.
    - Set action thresholds to determine when pest control measures should be taken based on the presence or level of pest activity.

- Regularly monitor and document pest activity through visual inspections, trapping, and use of pest monitoring devices.

3. Sanitation Practices:
   - Follow strict sanitation practices throughout the facility, including regular cleaning of equipment, floors, walls, and other surfaces to eliminate potential food sources for pests.
   - Properly dispose of waste and garbage in sealed containers to minimize pest attraction.
   - Ensure proper drainage and minimize standing water to prevent pest breeding sites.
4. Employee Education and Training:
   - Train employees on the importance of pest control, their role in preventing pest infestations, and the proper procedures for reporting and addressing pest-related issues.
   - Foster a culture of awareness and accountability among employees to promptly report any signs of pests or conditions that may contribute to pest problems.
5. Monitoring and Documentation:
   - Implement a systematic monitoring program to detect signs of pest activity, such as droppings, gnaw marks, or sightings, and record findings.
   - Maintain accurate records of pest control activities, including inspections, treatments, and corrective actions taken, as well as any recommendations provided by pest control professionals.
6. Collaboration with Pest Control Professionals:
   - Establish a partnership with a licensed pest control company to conduct regular inspections, provide expert advice, and implement targeted pest control measures.
   - Collaborate with pest control professionals to develop a customized pest control plan based on the specific needs and challenges of the food factory.
7. Regular Audits and Reviews:
   - Conduct regular internal audits and reviews of the in-process pest control program to assess its effectiveness, identify areas for improvement, and ensure compliance with regulations and industry standards.
8. Facility Maintenance and Structural Exclusion:
   - Regularly inspect the facility for cracks, gaps, and other potential entry points that pests can use to gain access. Seal or repair any openings in walls, floors, doors, windows, and utility penetrations.
   - Install door sweeps, weather stripping, and screens to prevent pests from entering through openings.
   - Maintain a well-maintained and clean facility to eliminate potential pest habitats and food sources.
9. Sanitation and Waste Management:
   - Establish and enforce strict sanitation protocols throughout the facility, including proper cleaning and removal of food debris, spills, and waste promptly.
   - Store and dispose of garbage in sealed containers and ensure proper waste management practices to reduce pest attraction.
10. Landscape and Exterior Management:
    - Maintain the surrounding landscape, ensuring that it is well-trimmed, free of overgrown vegetation, and kept away from the building to minimize pest harborage areas.
    - Properly store outdoor equipment, materials, and pallets away from the building to reduce potential pest habitats.
11. Storage and Product Handling:
    - Store raw materials, ingredients, and finished products off the floor and away from walls to facilitate cleaning and reduce pest access.

- Use proper storage containers and tightly sealed packaging to prevent pests from infesting stored products.
- Implement a first-in, first-out (FIFO) inventory system to minimize the storage time of materials and reduce the risk of pest infestation.

12. Regular Inspections and Monitoring:
    - Conduct regular inspections of the facility, both internally and externally, to identify any signs of pest activity or conditions that could attract pests.
    - Implement monitoring tools, such as insect traps, pheromone traps, or electronic monitoring systems, to detect early signs of pest presence and take timely action.
13. Pest Control Partnership:
    - Establish a relationship with a professional pest control provider who specializes in preventive strategies.
    - Collaborate with the pest control provider to develop and implement a customized preventive pest control program based on the specific needs and challenges of the food factory.

Corrective control refers to the actions taken to address and rectify issues related to pest infestations or other non-compliance with pest control measures. While preventive pest control focuses on proactive measures to prevent pests, corrective control comes into play when pests have already been detected or when there are breaches in the preventive measures. Here are some key aspects of corrective control:

1. Pest Identification and Assessment:
    - Properly identify the pests present in the facility to determine the extent of the infestation and potential risks associated with them.
    - Assess the areas affected by the infestation, including identifying the source of the problem and evaluating the extent of damage or contamination caused.
2. Immediate Action:
    - Take prompt action to eliminate the pests and address the specific issues identified during the assessment.
    - Depending on the severity of the infestation, this may involve measures such as targeted pesticide application, trapping, removal of pest habitats, or physical removal of pests.
3. Root Cause Analysis:
    - Conduct a thorough investigation to identify the underlying causes that allowed the pest infestation to occur.
    - Determine if there were any gaps or failures in the preventive pest control measures, such as structural deficiencies, sanitation lapses, or employee practices.
4. Corrective Measures and Remediation:
    - Develop and implement corrective actions to address the identified root causes and prevent future pest infestations.
    - This may include repairing structural issues, improving sanitation protocols, enhancing employee training, or modifying processes to eliminate pest-attracting conditions.
5. Documentation and Record-Keeping:
    - Maintain accurate records of the pest infestation incident, actions taken, and the effectiveness of corrective measures.
    - Documentation should include details of pest control treatments, inspection reports, employee training records, and any changes made to preventive measures.
6. Monitoring and Follow-up:
    - Regularly monitor and inspect the facility to ensure that the corrective actions have effectively resolved the pest issue.

- Implement follow-up inspections to verify that the identified root causes have been addressed and that preventive measures are being properly implemented.

7. Continuous Improvement:
   - Use the pest infestation incident as an opportunity for learning and continuous improvement.
   - Review and update preventive pest control protocols, employee training programs, and facility maintenance practices to prevent similar issues in the future.

4.25. FUMIGAGTION AND PESTICIDE CONTROL IN PROCESS ENVIRONMENT

Fumigation and pesticide control are two methods used in pest management to control and eliminate pests in food factories. However, it's important to note that the use of fumigation and pesticides should be carefully considered and conducted by trained professionals following regulatory guidelines and best practices. Fumigation is a method of pest control that involves the use of gaseous pesticides, known as fumigants, to eliminate pests in an enclosed space or within certain commodities. It is typically used to control pests such as stored product insects, rodents, and insects that infest structures or equipment.

- Fumigation requires specialized knowledge and equipment, and it is usually performed by licensed professionals.
- The process involves sealing the area or commodity to be fumigated, introducing the fumigant gas, and allowing sufficient time for the gas to penetrate and kill the pests.
- Fumigated areas or commodities need to be properly aerated to remove the fumigant gas residue and ensure it is safe for workers and consumers.

Pesticides are chemical substances designed to control or eliminate pests, including insects, rodents, and other pests that can pose a threat to food safety and product quality.

- Pesticide control in food factories involves the strategic application of approved pesticides to target pests and minimize their impact on the facility and products.
- It is crucial to use pesticides approved for use in food processing facilities and comply with regulatory requirements regarding their application and usage.
- Pesticides should only be applied by trained and licensed personnel following label instructions, safety precautions, and appropriate application methods.
- Integrated Pest Management (IPM) principles should be considered to minimize reliance on pesticides and employ a comprehensive approach that includes prevention, monitoring, and non-chemical control methods.

4.26. PHYSICAL CONTROL – BUILDING AND TRAP MAPS

Physical and building controls are important components of a comprehensive pest control program in OFSP processing factories. These controls involve implementing measures and practices that prevent pests from entering the facility or accessing food storage and processing areas. Here are some key aspects of physical and building controls for pest management:

1. Facility Design and Construction:
   - Design and construct the facility with pest control in mind, incorporating features that minimize potential entry points for pests.
   - Ensure that walls, floors, and ceilings are constructed with materials that are resistant to pest intrusion and easy to clean.
   - Install tight-fitting doors and windows and use screens to prevent pests from entering through openings.

2. Entry Point Management
    - Seal or screen all openings, gaps, and cracks in the building structure, including utility penetrations, vents, and piping.
    - Install door sweeps, weather stripping, and air curtains to prevent pests from entering through doors and dock areas.
    - Regularly inspect and maintain exterior areas, including loading docks, to prevent pest entry.
3. Ventilation and Airflow Control:
    - Design and maintain a ventilation system that minimizes the entry of pests.
    - Use screens or filters on air intakes to prevent insects and other pests from entering through HVAC systems.
    - Ensure proper airflow in food processing areas to prevent the accumulation of odors or food particles that can attract pests.
4. Drainage Management:
    - Maintain proper drainage systems to prevent the accumulation of water, which can attract pests.
    - Regularly clean and inspect drains to remove debris and organic matter that can serve as food sources for pests.
    - Install drain covers or screens to prevent pests from entering through drains.
5. Lighting Management:
    - Use lighting fixtures that are designed to minimize attracting pests, such as insects.
    - Position lights away from entrances or areas where pests can be attracted to the light source.
6. Waste Management:
    - Implement effective waste management practices, including proper storage, disposal, and cleaning of waste containers.
    - Keep waste storage areas clean, secure, and located away from the main facility to minimize pest attraction.
7. Storage and Shelving:
    - Store materials and products off the floor and away from walls on clean, well-maintained shelving.
    - Use pallets or racks with gaps to facilitate inspection and cleaning.
    - Regularly inspect and clean storage areas to prevent pest harborage.
8. Pest Monitoring Devices:
    - Install and regularly monitor pest control devices, such as insect traps or pheromone monitors, to detect and monitor pest activity in critical areas.
    - Analyze the data from monitoring devices to identify trends and take appropriate preventive actions.
9. Employee Education and Training:
    - Train employees on the importance of physical and building controls for pest management.
    - Educate them on proper sanitation practices, waste management, and their role in reporting and addressing potential pest entry points.

MODULE 6: PERSONNEL

4.27. MINIMUM REQUITMENTS TO WORK IN FOOD FACTORY

Any personnel working in a food factory must have the necessary education, training, or experience to do so.

- Having a biological science degree was seen as enough in the time past
- A new law was signed in 2019 for the establishment of a Nigerian Council for Food Science and Technologists. This may require being licensed to work in the Food industry soon.
- Personnel that will handle food must be healthy and whole with no open sore or communicable disease.

- Personnel working in the food processing areas must understand the process flows and the Standard Operating Procedures (Training and Experience can achieve this).

4.28. FOOD HANDLERS TEST

A food handler is any personnel in the food industry who is involved in processing and packaging of the product directly. personnel such as drivers and cleaners as well as other ad-hoc staff may not be classified as food handlers if they have no access to the processed food or primary packaging material

- A food handlers test is expected to be carried out biannually (Every six months) to confirm that food handlers do not carry any pathogen that can be transmitted to consumers through the food they handle
- A Certified Medical laboratory is expected to carry out the test or the Hygiene department of the Ministry of Health in Nigeria also carries out this test (either of the two results are acceptable)
- NAFDAC, SON, Ministry of Health and other regulators will request for this per time
- It is also a requirement for food registration and certification

4.29. PRINCIPLE REGARDING SORE, NURSING MOTHERS, AND CROSS CONTAMINATION

Personnel in the food processing area have the potential to contaminate products through external contacts or personal habits within the production environment.

- Contamination can be from external contacts or habits inside the production Environment
- Contamination can also be from poor maintenance, poor cleaning, or chemical residue from activities such as fumigation.
- The personnel must use all hygiene and sanitation points within the production area when necessary
- They are not allowed to bring any item (including food material) not involved in production into the food processing area
- They are expected to put on Protective clothing that is meant for the production areas only
- Protective Equipment should be used based on the requirement of production environment
- PPE that is common to use will include gloves, nose masks, Protective foot covering, head covering as well as eye googles.
- Access control to ensure personnel who are not food handlers do not have access to the processing area.
- protective clothing should not be taken outside the production premises (cloak room for changing should be available)
- PPE being used must be always clean.

4.30. SUITABLE USE OF PPE WITHIN PLANT

The use of personal protective equipment (PPE) is essential in the production area of a vitamin A cassava processing facility to protect workers from potential hazards and ensure their safety. PPEs commonly used in the production area:

1. Safety Helmets/Hard Hats: Safety helmets or hard hats are worn to protect the head from falling objects, impact hazards, or overhead structures. They provide protection against potential head injuries in areas where there is a risk of falling debris or equipment.
2. Safety Glasses/Goggles: Safety glasses or goggles are used to protect the eyes from dust, flying particles, chemicals, or other hazards that may cause eye injuries. They should have impact resistance and may also offer protection against liquid splashes or harmful vapors.

3. Face Shields: Face shields provide full-face protection and are typically used in conjunction with safety glasses or goggles. They shield the face from impact hazards, splashes, sparks, or chemical exposure. Face shields are commonly used when working with machinery, grinding, or handling hazardous substances.
4. Respiratory Protection: Depending on the specific tasks and potential exposure to airborne contaminants, respiratory protection may be required. This can include dust masks, particulate respirators, or air-purifying respirators. These devices protect the respiratory system from inhaling dust, fumes, gases, or harmful substances.
5. Protective Clothing/Coveralls: Protective clothing, such as coveralls or disposable suits, is worn to protect the body from direct contact with hazardous materials, chemicals, or biological agents. They should be made of durable and chemical-resistant materials, covering the entire body, and providing an extra layer of protection.
6. Gloves: Depending on the tasks involved, workers may require different types of gloves. For example, cut-resistant gloves protect against sharp objects, heat-resistant gloves provide protection from high temperatures, and chemical-resistant gloves protect against harmful substances. Gloves should be selected based on the specific hazards present in the production area.
7. Hearing Protection: In areas with high noise levels, such as machinery operation or grain processing, hearing protection should be provided. This can include earplugs or earmuffs designed to reduce the risk of noise-induced hearing loss.
8. Safety Footwear: Safety footwear, such as steel-toe boots, is worn to protect the feet from heavy objects, falling objects, or crushing hazards. They should have slip-resistant soles and provide adequate support and protection for the feet.
9. Aprons and Sleeves: Aprons and sleeves made of durable materials can provide protection against chemical splashes, hot liquids, or other liquid hazards. They are commonly used when handling chemicals, cleaning equipment, or working with hot liquids.
10. Fall Protection Equipment: In areas where there is a risk of falls from heights, fall protection equipment such as safety harnesses, lanyards, and anchor points should be provided to prevent workers from falling and sustaining serious injuries.

4.31. KNOWLEDGE UPDATE

Training and experience are important for any industry to survive

- Experience can lead to oversight and trivializing of important processes by long standing staff of the company (familiarity to processes may make experienced staff to overlook food safety risk)
- Constant training and monitoring of personnel is important to ensure processes are followed as required
- When there is a change in process, equipment, legislation, regulatory requirement, or any changes that may affect production (including consumer change in preference), Personnel should undergo a knowledge update training to bring them to speed on these developments
- When a personnel is being introduced to a new section or changing roles, proper induction training should be done even if they are familiar with the operations of such section.
- Organization can develop a routine training/Awareness plan that should be strictly followed to ensure personnel are aware of requirements and updated all the time
- Resources should be allocated to training by the Management either through budget or through approved expenses

4.32. PERSONAL HYGINENE

Personnel should be trained on Hygiene requirement during induction or before they join the team

- Ensure each personnel goes through the Food handlers test before they join the team

- Discourage unhealthy practices such as using fingers to pick the nostrils
- Personnel Hair should be short and covered
- Personnel should not wear flowing jewelries especially around machines
- Personnel should have short nails that cannot harbour dirt
- Personnel should learn to limit handshakes, touching of surfaces or contact with machine parts that can lead to product contamination
- Personnel should wash their hands before entering production premises, when necessary, within production premises and after they have contact with surfaces or machine parts that can contaminate production premises.
- No Use of excessive Make up when coming to work in the factory premises
- Lactating mothers can be restricted to non-production areas except when hygiene can be ascertained.
- No Personnel with open wound should be allowed into Production areas
- Personnel with symptoms of coughing, sneezing, drooling nose, red eyes or fatigue should be immediately exempted from production areas
- Personnel with dysentery, or with evidence of "running stomach" should be exempted from production premises immediately

5. FINISHED PRODUCT HANDLING: STORAGE AND PACKAGING TO MEET REGULATORY STANDARDS

MODULE 1: PRODUCT-SPECIFIC REQUIREMENTS

5.1. INTRODUCTION TO STANDARDS

Standards are consensus documents used to determine the criteria for products and services. It involves all stakeholders within an industry including consumers, regulators, and industry players to agree on the criteria for a product. All Standards in Nigeria are in custody of Standards Organization of Nigeria as a secretariat. Standards in Nigeria are mandatory and requires full implementation by all stakeholders.

Codex is the International Organization in charge of Food Standardization. Codex has developed limit for important limits such as heavy metal, aflatoxin, and codes of hygiene for food product environment. These standards are fundamentals for food factory set up. Each product has a standard that specifies the requirement of the product. These requirements are general as recommended by Codex Standard.

5.2. SPECIFIC REQUIRMENT LIMITS AND NATIONAL STANDARDS FOR QUALITY ASSURANCE

Product standard highlights the specific requirement for each product. Specific requirement may include PH range, acidity, moisture, microbial load, Heavy metal limit, pesticide limit and Aflatoxin limit. Specifically, for Biofortified Cassava and all Cassava products, level of cyanogenic compounds in the cassava must be within limit. Heavy metals and aflatoxin requirements in rice have strict limit. Each specification should be followed as stated in the National Industrial Standards (NIS). General codex limits for Heavy Metals and aflatoxin usually applies for the different food products.

S/NO	PARAMETERS	CONSIDERED LIMIT

1	Moisture	% should not be more than 10 in a dry product
2	Ash Content	Depends on type and variety of Products
3	Crude fiber	Depends on the type and Variety of Products
4	Hydrogen Cyanide	0.01mg/100g
5	Protein	Depends on type and variety of Products
6	Carbohydrate as Sugar	Depends on type and Variety of Product
7	PH	Depends on type of Product- lower for fermented products

S/NO	PARAMETER	GENERAL EXPECTED VALUE (CFU)
01	Total Viable Count	1000
02	Escherichia Coli	Nil
03	Coliform	Nil
04	S. Aureus	Nil
05	Salmonella spp.	Nil
06	Yeast and Mold	100

5.3. NATIONAL CERTIFICATION

National Food and Drug Administration and Control (NAFDAC)

Food Products are to be registered with the National Food and Drug Administration and Control (NAFDAC). The procedure includes inspection, testing, ensuring personnel are competent before registration is granted. A registered number unique to a product is released for the product to be used. The number/registration is renewed every five years subject to confirmation of product requirements and positive inspection feedback. NAFDAC will also monitor the distribution and sales of the products and can conduct market survey of product within the five years. NAFDAC is responsible for regulating food products adverts in Nigeria and all adverts for food products are expected to be approved by them first. NAFDAC is a member of the National Codex committee and works with other members to contribute to International Food Standardization.

Standards Organization of Nigeria (SON)

Standards Organization of Nigeria (SON) is responsible for the drafting of National Standards and as Secretariat for same. SON is the Secretariat for Codex activities in Nigeria and works with other agencies to arrive at a position for Nigeria in International Food Standardization. SON enforces National Standard under the Mandatory Conformity Assessment Programme (MANCAP) Scheme. SON offers Certification using the national or relevant international standards to determine conformity. SON certification includes testing, inspection, third party testing and quarterly inspection of facility. This also includes machinery, arrangement, preventive maintenance, and inventory. SON certification allows Nigerian products to be accepted in Almost all African Countries without query. SON Certification is more cumbersome to achieve and is valid for only three years before renewal. SON will carry out quarterly inspection and testing of products under certification. The Federal and State Ministry of Health are involved in facility and personnel conformance. The unit involved in food factory inspection monitors the hygiene of production environment as well as personnel hygiene. They also regulate the environment and pest control of the facility

National Quarantine Service

The National Quarantine service under the Ministry of Agriculture monitors food and agricultural produce conformity. They issue Phytosanitary license to products for exports. This requirement allows a lot of Nigerian produce to gain acceptance to other countries.

5.4. NUTRITIVE COMPONENT PRESERVATION

Major food nutritive components can be destroyed during processing. Example of processing and nutritive loss is the leaching of Vitamin C from wastewater in food. Vitamin A is also light sensitive and can easily be degraded upon excess exposure to light.

Other food nutritive components such as proteins and minerals are sensitive to heat, fermentation, acidic content, and microbial action. Controlled fermentation in a suitable environment has been established to not only preserve the nutrient but also to enhance it

Drying to a low moisture content will also reduce the possibility of microbial activity and nutrient loss. Ensuring that heat, light and moisture regulation is monitored to avoid nutritive loss. Over pressing and leaching should also be minimized.

5.5. WATER TREATMENT AND USAGE

Water to be used in cassava food processing must be safe and not contaminate the product. Water treatment must be taken as a complete process on its own. Water treatment required will include filtration, sanitation, removal of chemical contaminants as well as removal of taste odor and color.

Use sand filters, carbon filter, UV sterilizer, micron filters as well as antimicrobial agent to remove all microbial load. Storage tanks should be disinfected and cleaned properly to avoid contamination of already treated water

Quality assurance testing should be carried out on all raw water and finished water to be used for production. Result of testing should be available towards regulatory requirements.

5.6. THIRD-PARTY TESTING – INTRODUCTION TO IPAN

Third party analysis is carried out by an independent laboratory using Regulatory and National Standard Criteria. It is called comprehensive analysis because it covers all requirements stated in the standard beyond what the in-house capacity can test.

Product testing can be carried out in any laboratory as far as they meet necessary requirement. For local testing, ensure that the product is tested in a laboratory with at least one IPAN certified personnel. For international testing, ensure that you test in a Laboratory with an ISO 17025 certification for specific parameters.

The Institute of Public Analyst of Nigeria (IPAN) is the body responsible for regulating the testing of chemical and food products in Nigeria. The Institute regulates the personnel as well as laboratory functionality. Results of analysis from any laboratory with IPAN personnel is expected to carry the seal of the Institute. Regulatory agencies may reject Laboratory results that does not carry the seal of the Institute.

5.7. RELEASE OF PRODUCT – CRITERIA AND METHOD (BATCHING)

Ensure product batching is done using characteristics of the raw material, date and time of production and other relevant information. Product batch should be labeled accordingly and identifiable both by personnel and the customer. Each batch should be tested to ensure they meet physical, chemical, and microbial requirement before they are passed for release.

A dedicated personnel must be responsible for final release of product before it is allowed for sales. A dedicated record must be available for the evidence of release of product. Reference samples should be kept for each batch of released product to serve as reference in cases of complaint or recall

Products that fail to meet release criteria should be treated as nonconforming product. A product that failed critical parameters should either be reworked or discarded. If reworked, the approved authority required to allow rework should be clear and records of such presented to regulatory authorities as necessary.

Common criteria for release of product include:

- Sealing integrity
- Moisture content
- PH
- Microbial load-E.Coli, Salmonella and Total Viable Count
- Chemical requirement
- Acceptable colour (for rice)
- Acceptable odour and flavour (especially for fermented foods)
- Evidence of batch codes on final products
- Weight and volume accuracy

MODULE 2: PRODUCT PACKAGING

5.8. SELECTION OF PACKAGING

When it comes to selecting packaging, there are several factors to consider, including the nature of the product, its requirements, branding, sustainability, and cost. Some considerations for selecting packaging are

1. Product Protection: The primary function of packaging is to protect the product during storage, transportation, and handling. Consider the fragility, perishability, and sensitivity of the product to determine the appropriate packaging material and design.
2. Packaging Material: Choose packaging materials that are suitable for the product and align with your sustainability goals. Common options include cardboard, paperboard, plastic, glass, metal, or a combination thereof. Consider factors like durability, recyclability, biodegradability, and composability.
3. Size and Shape: The packaging should be appropriate in size and shape to accommodate the product effectively. Oversized packaging can lead to excessive material waste and increased shipping costs, while undersized packaging may not provide adequate protection.

4. Branding and Design: Packaging plays a crucial role in conveying your brand's identity and attracting consumers. Consider incorporating your brand's colors, logo, and messaging into the packaging design to enhance recognition and create a memorable experience.
5. Regulatory Compliance: Ensure that the packaging complies with relevant industry regulations, safety standards, and labeling requirements. Depending on the product, you may need to meet specific guidelines for food, pharmaceuticals, hazardous materials, or other regulated items.
6. Sustainability: Increasingly, consumers are looking for environmentally friendly packaging options. Consider using recyclable, biodegradable, or compostable materials. You can also explore innovative packaging solutions like eco-friendly alternatives to plastic or reduced packaging waste
7. Cost: Packaging costs can significantly impact your overall product expenses. Evaluate the cost-effectiveness of different packaging options while considering factors like material, production, transportation, and storage. Balance your budget with the desired level of quality and functionality.
8. Consumer Convenience: Consider the ease of use and convenience for the end consumer. Packaging should be user-friendly, easy to open, and provide relevant information such as instructions, ingredients, and nutritional facts.
9. Supply Chain Considerations: Assess the entire supply chain when selecting packaging. Consider factors like compatibility with automated packaging systems, storage requirements, transportation efficiency, and any special handling or labeling requirements.
10. Market Trends: Stay updated on the latest packaging trends and innovations in your industry. This includes emerging technologies, smart packaging, interactive designs, or any other features that can enhance the consumer experience.

5.9. STORAGE OF PACKAGING MATERIALS

When it comes to selecting storage packaging materials, the primary considerations are durability, protection, and organization. Some commonly used storage packaging materials:

1. Cardboard Boxes: Cardboard boxes are versatile, affordable, and widely used for storage purposes. They come in various sizes and strengths to accommodate different items. They provide good protection against light impacts and can be stacked for efficient use of space.
2. Plastic Bins and Totes: Plastic bins and totes are durable and offer better protection against moisture and pests compared to cardboard. They are available in different sizes and styles, often with lids that provide secure closure. Plastic bins are reusable and can be stacked to optimize storage space.
3. Shelving Units: While not strictly packaging materials, shelving units are essential for organizing and storing items. They can be made of materials like metal, wood, or plastic. Shelving units provide efficient vertical storage and easy access to items.
4. Bubble Wrap and Foam: For delicate or fragile items, bubble wrap or foam sheets provide cushioning and protection. They are used to wrap individual items or line the inside of boxes to absorb shocks and prevent damage during storage and transportation.
5. Plastic Wrap and Stretch Film: Plastic wrap or stretch film is useful for bundling or wrapping multiple items together. It secures loose items, prevents dust accumulation, and offers some level of protection against moisture and dirt.
6. Pallets: Pallets are used for storing and transporting heavy or bulky items. They provide a stable base and make it easier to move items using forklifts or pallet jacks. Pallets can be made of wood, plastic, or metal, and they are often combined with stretch wrap for secure packaging.

7. Vacuum-sealed Bags: Vacuum-sealed bags are useful for storing clothing, bedding, and other textiles. They remove air from the bags, reducing bulk and providing protection against moisture, dust, and pests. Vacuum-sealed bags can save storage space and help keep items organized.
8. Corrugated Plastic Sheets: Corrugated plastic sheets, also known as Coroplast, are lightweight, durable, and water-resistant. They are often used as dividers, partitions, or protective layers between items during storage. Corrugated plastic sheets are reusable and can be easily cut to size.
9. Labels and Markers: Labels and markers are crucial for organizing and identifying stored items. They help you quickly locate specific items and maintain an organized storage system. Use waterproof markers and adhesive labels for durability.

IMAGE COLLAGE OF SUITABLE VIT. A CASSAVA PACKAGING MATERIALS

5.10. PACKAGING ACCORDING TO PRODUCT CHARECTERISTICS

When selecting packaging according to product characteristics, it's important to consider the specific needs and attributes of the product. Some common product characteristics and corresponding packaging considerations are:

1. Fragile Products: Fragile items require packaging that provides excellent protection against impacts and shocks. Consider using packaging materials with cushioning properties such as foam inserts, bubble wrap, or air-filled packaging. Use sturdy boxes or containers that can withstand external pressure and provide structural support.
2. Perishable or Food Products: Perishable or food items require packaging that ensures freshness, prevents contamination, and extends shelf life. Consider using airtight containers, vacuum-sealed bags, or specialized food packaging with features like moisture barriers or temperature control. Label the packaging with clear expiration dates and handling instructions.
3. Hazardous or Chemical Products: Hazardous or chemical products must comply with strict safety regulations. Use packaging materials that are compatible with the product and provide containment to prevent leaks or spills. Consider using specialized containers, labels, and warning signs that comply with applicable regulations for hazardous materials.
4. Temperature-Sensitive Products: Temperature-sensitive items, such as pharmaceuticals or certain food products, may require packaging that maintains specific temperature ranges. Insulated packaging, such as thermal containers or coolers, can help regulate temperature during storage and transportation. Consider using gel packs, dry ice, or other temperature-controlling materials.
5. Liquid or Fluid Products: Packaging for liquid or fluid products should be leak-proof and resistant to breakage. Bottles, jars, or containers made from durable materials like glass or plastic with tight-sealing caps are commonly used. Consider additional measures such as tamper-evident seals or protective secondary packaging to ensure product integrity.
6. Oddly Shaped or Bulky Products: For irregularly shaped or bulky items, custom packaging solutions may be necessary. Consider using custom-made boxes, foam inserts, or molded packaging that conform to the shape of the product. This helps provide secure protection and prevents movement during handling and transport.
7. Small or Delicate Products: Small or delicate products may require packaging that provides individual protection and prevents damage. Consider using small boxes, blister packs, or clamshell packaging that securely holds each item. Cushioning materials like foam, tissue paper, or cardboard dividers can prevent scratches or breakage.
8. High-End or Luxury Products: Luxury products often require packaging that reflects their premium status and enhances the customer experience. Consider using high-quality materials, elegant designs, and custom branding elements. Premium boxes, specialized inserts, and decorative finishes can create a luxurious presentation.
9. Eco-Friendly or Sustainable Products: If your product aligns with eco-friendly values, choose packaging materials that are recyclable, biodegradable, or made from renewable resources. Consider using minimal packaging, eliminating excess materials, or incorporating eco-friendly printing inks and labels.

It is also important to consider the following:

- Product branding: The packaging should be designed to reflect the brand of the product.
- Product marketing: The packaging should be designed to promote the product to consumers.
- Product safety: The packaging should be designed to protect the product from contamination and spoilage.
- Product environmental impact: The packaging should be designed to have a minimal environmental impact

5.11. PACKAGING AS ADVERTORIAL

Packages can serve as advertorial tools to boost brand recognition, create a memorable unboxing experience, and engage customers. Here are some approaches to using packages as advertorials:

1. Branding Elements: Incorporate your brand's logo, colors, and tagline on the packaging to reinforce brand identity and ensure consistency with your visual identity.

2. Customization: Design unique packaging that reflects your brand or product's distinctiveness. This can include innovative shapes, personalized messages, or custom prints to make a lasting impression.

3. Storytelling: Utilize packaging to tell your brand's story, values, or the narrative behind the product. Consider adding informative inserts, product guides, or personal letters to foster an emotional connection with customers.

4. Unboxing Experience: Create packaging that offers an exciting unboxing experience. This can involve multiple layers, hidden surprises, or interactive elements that captivate customers during the unwrapping process.

5. Samples and Promotions: Include product samples, promotional items, or exclusive offers within the package to incentivize repeat purchases and cultivate customer loyalty.

6. QR Codes or NFC Tags: Incorporate QR codes or NFC tags on the packaging for customers to scan with their smartphones. These can be linked to interactive content, special offers, or additional product information, enhancing customer engagement.

7. Social Media Integration: Encourage customers to share their unboxing experiences on social media by incorporating relevant hashtags, handles, or instructions within the packaging. This can expand your brand's reach and generate user-generated content.

8. Sustainable Packaging: Demonstrate your commitment to sustainability by utilizing eco-friendly packaging materials and highlighting their environmental benefits, such as recyclability or renewable sources.

9. Personalized Touches: Add a personal touch to the packaging by including handwritten thank-you notes, customized stickers, or personalized packaging inserts. This creates an authentic connection with customers and shows appreciation for their support.

Customer Feedback: Include a feedback card or QR code to encourage customers to provide feedback or reviews about their experience with the product and packaging. This helps gather valuable insights, improve customer satisfaction, and enhance your brand's reputation.

5.12. HERMETIC PACKAGING AND BREATHABLE PACKAGING REQUIRMENTS

Hermetic packaging refers to a type of packaging that is completely sealed and airtight, providing a barrier against the exchange of gases or moisture between the inside and outside of the package. On the other hand, breathable packaging materials allow for controlled exchange of gases or moisture, typically designed to maintain the freshness or stability of certain products. Here's an explanation of both hermetic and breathable packaging materials:

Hermetic Packaging Material

Hermetic packaging materials are impermeable to gases and moisture. They create a complete barrier that prevents the exchange of air, moisture, or other contaminants. Hermetic packaging is commonly used for products that require protection from external factors, such as sensitive food products like flowers, ready to eat snacks and so on.

These materials ensure that the product remains protected from oxygen, moisture, and other potentially damaging elements, extending its shelf life and maintaining its quality.

Breathable Packaging Material

Breathable packaging materials are designed to allow controlled exchange of gases or moisture. These materials are typically used for products that require specific gas permeability or moisture regulation to maintain freshness or stability. Examples of products that benefit from breathable packaging include certain types of fresh produce, flowers, and some pharmaceuticals.

Breathable packaging materials may include:

1. Perforated films or pouches: Films or pouches with small holes or perforations that allow for controlled gas exchange.

2. Micro-perforated films: Films with very tiny perforations that enable controlled airflow while still providing a barrier against contaminants.

3. Membranes or specialized coatings: These materials can be applied to packaging to regulate moisture or gas exchange based on specific product requirements.

4. Breathable packaging materials help regulate the internal environment of the package, allowing for the release of excess moisture or gases while preventing external contaminants from entering. This helps maintain the quality, freshness, and shelf life of the packaged product.

5.13. PACKAGING AGAINST PEST CONTROL

Packaging plays a crucial role in protecting products from pests during storage and transportation. Here are some strategies for packaging against pest control:

1. Choose Pest-Resistant Materials: Select packaging materials that are less susceptible to damage from pests. For example, consider using sturdy materials such as thick cardboard, plastic, or metal that are less prone to infestation or penetration by pests.
2. Seal Packaging Securely: Ensure that packaging is tightly sealed to prevent pests from accessing the product. Use adhesive tapes, straps, or secure closures to create a barrier that pests cannot easily breach.
3. Implement Barriers: Introduce additional barriers to deter or prevent pests from reaching the product. This could include using shrink wrap, plastic bags, or airtight containers to create an extra layer of protection.
4. Consider Insect-Repellent Materials: Incorporate insect-repellent materials or coatings into the packaging to discourage pests. Certain materials, such as insect-repellent films or treated papers, can help repel or inhibit the presence of pests.
5. Proper Storage and Handling: Store packaged products in areas that are clean, dry, and well-ventilated. Keep the storage facility free from debris, food residue, and other factors that could attract pests. Implement good inventory management practices to rotate products and prevent prolonged storage.

6. Regular Inspection and Maintenance: Conduct routine inspections of packaging and storage areas to identify signs of pest activity. Look for holes, gnaw marks, or any indications of infestation. If pests are detected, take appropriate action to eradicate them and repair or replace damaged packaging.
7. Implement Pest Control Measures: Employ preventive pest control measures in the storage facility to minimize the risk of infestation. This can include regular cleaning, pest monitoring, use of traps or baits, and, if necessary, the application of approved pesticides by trained professionals.
8. Educate and Train Staff: Provide training to employees involved in packaging and storage processes on proper pest control practices. Educate them about early detection, reporting procedures, and the importance of maintaining a pest-free environment.
9. Traceability and Quality Control: Implement traceability systems to identify and track any potential sources of pest infestation. Establish quality control processes to ensure that incoming products or materials are free from pests before packaging.
10. Collaborate with Pest Control Professionals: Seek advice from pest control experts or engage professional pest control services to develop a comprehensive pest management plan tailored to your specific packaging and storage needs.

5.14. STANDARDS REQUIRMENTS ON PACKAGING

The Nigerian Standard Organization (SON) is responsible for establishing and enforcing product packaging regulations in Nigeria. It is crucial to consult the Nigerian Standard Organization (SON) or seek legal advice to ensure compliance with the specific packaging requirements for your product category. Regulations may evolve, so staying updated with the latest guidelines and standards is essential to ensure compliance with Nigerian packaging regulations. While specific requirements may vary depending on the product category, there are general standards that businesses should adhere to when packaging their products in Nigeria. Some key considerations are:

1. Labeling Requirements: Products should be labeled in accordance with SON's guidelines. The label should include essential information such as product name, manufacturer's name and address, batch/lot number, manufacturing/expiry dates (where applicable), net weight or volume, ingredients (if applicable), usage instructions, warnings, and any other information required by specific regulations.

2. Language: All labeling, including text, instructions, and warnings, should be in English or a language that consumers can easily understand.

3. Nutritional Information (if applicable): Food products should display accurate nutritional information, including the composition of ingredients, nutritional values, and allergen information, as required by SON's guidelines.

4. Packaging Material Safety: Packaging materials should comply with relevant safety standards to ensure that they do not pose a risk to the product or consumer's health. This includes ensuring that packaging materials are free from contaminants and do not transfer harmful substances to the product.

5. Weight and Volume Accuracy: Products sold by weight or volume should comply with SON's guidelines regarding accuracy in measurement. It is important to use calibrated weighing and measuring equipment to ensure the declared weight or volume is correct.

6. Child-Resistant Packaging (if applicable): Certain products, such as medications or household chemicals, may require child-resistant packaging to reduce the risk of accidental ingestion. Packaging for such products should meet the relevant child-resistant packaging standards.

7. Packaging Material Recycling and Disposal: Consideration should be given to the recyclability and environmental impact of packaging materials. It is advisable to use packaging materials that are recyclable or environmentally friendly and provide appropriate disposal instructions on the packaging.

8. Packaging Stability and Durability: Packaging should be designed to protect the product during handling, storage, and transportation. It should be robust enough to withstand foreseeable stresses and environmental conditions to maintain the quality and integrity of the product.

9. Packaging Graphics and Branding: Any graphics, logos, or branding elements on the packaging should comply with applicable trademark and copyright laws. It is important to avoid misleading or deceptive packaging that may misrepresent the product or confuse consumers.

5.15. LABELING AND DATE CODING

Labelling and coding are essential aspects of product packaging that provide important information to consumers, facilitate traceability, and ensure regulatory compliance. Here's an overview of labelling and coding considerations:

1. Mandatory Labeling Information:
    - Product Name: Clearly indicate the name of the product on the label.
    - Manufacturer Information: Include the name and address of the manufacturer, packager, or distributor.
    - Net Quantity: Provide the accurate net weight, volume, or count of the product.
    - Ingredients: If applicable, list all ingredients or components used in the product.
    - Allergen Information: Highlight any potential allergens present in the product.
    - Batch/Lot Number: Assign a unique identifier to each batch or lot for traceability.
    - Manufacturing/Expiry Dates: Clearly display the manufacturing date and, if applicable, the expiration or best before date.
    - Usage Instructions: Provide clear instructions for proper use, storage, or handling of the product.
    - Warnings and Precautions: Include any necessary warnings, precautions, or usage restrictions for consumer safety.
    - Country of Origin: Specify the country of origin for the product.
    - Barcodes: Include barcodes for easy scanning and inventory management.
2. Language and Legibility:
    - Use clear and legible fonts to ensure that the information on the label is easily readable.
    - Labels should be in a language that the target market can understand, typically in English or the official language of the country.
3. Label Placement and Size:
    - Labels should be prominently displayed on the product packaging, ensuring visibility and legibility for consumers.
    - Ensure that the label size is appropriate to accommodate all mandatory information without compromising readability.
4. Labeling Regulations and Standards:
    - Familiarize yourself with the specific labeling regulations and standards applicable to your product category, which may vary depending on the industry and country.
    - Adhere to any specific requirements related to font size, color, positioning, or information placement.
5. Coding and Traceability:
    - Implement coding systems, such as batch or lot numbers, to enable traceability and recall management.
    - Codes should be clear, durable, and easily readable, typically printed directly on the packaging or via labels or tags.
6. Regulatory Compliance:
    - Ensure compliance with relevant local and international regulations, such as those set by government agencies or industry-specific standards organizations.
7. Design and Branding:

- Consider the overall design and branding elements of the label to align with your brand identity and enhance product appeal.

8. Update and Review:
   - Regularly review and update labels to comply with any changes in regulations or standards.
   - It is important to consult applicable regulatory bodies, industry associations, or legal professionals to ensure compliance with specific labelling and coding requirements for your product category and target market.

5.16. PRIMARY AND SECONDARY PACKAGING REQUIREMENTS

PRIMARY PACKAGING: Primary packaging refers to the immediate packaging that directly contains the product. It is the layer of packaging that is in direct contact with the product itself. The specific requirements for primary packaging can vary depending on the nature of the product and the industry. However, here are some common considerations for primary packaging requirements:

1. Product Protection: The primary packaging should effectively protect the product from physical, chemical, and biological damage. It should prevent contamination, moisture ingress, light exposure, and other factors that could compromise the quality, safety, or efficacy of the product.
2. Material Compatibility: The materials used for primary packaging should be compatible with the product and not react or interact with it in a way that could cause harm or degradation. This is particularly important for food, pharmaceuticals, and other sensitive products.
3. Safety and Regulatory Compliance: Primary packaging should comply with relevant safety and regulatory standards specific to the product category. This includes considerations such as child-resistant packaging for certain medications or safety seals for food and beverages.
4. Quality and Durability: Primary packaging should be of high quality and designed to withstand handling, transportation, and storage conditions without being easily damaged. It should maintain its integrity throughout the product's shelf life.
5. Convenience and User-Friendliness: Primary packaging should be designed with user convenience in mind. It should be easy to open, close, dispense, or use, ensuring a positive user experience.
6. Branding and Product Presentation: Primary packaging often serves as a marketing tool and should be designed to effectively represent the brand and attract consumers. It should reflect the brand's identity, communicate product information, and enhance the visual appeal of the product.
7. Labeling and Information Display: Primary packaging should provide sufficient space for required labeling information, such as product name, manufacturer details, ingredients, usage instructions, and any regulatory information.
8. Environmental Considerations: Primary packaging should be designed with sustainability in mind, considering the use of eco-friendly materials and minimizing waste generation. It should align with relevant environmental standards and regulations.
9. Manufacturing Efficiency: Primary packaging should be designed for efficient and cost-effective manufacturing processes, considering factors such as ease of filling, sealing, labeling, and integration with automated packaging systems.
10. 10.Transportation and Storage Considerations: Primary packaging should be designed to facilitate easy stacking, handling, and transportation. It should also consider storage requirements, such as stackability or space optimization.

SECONDARY PACKAGING refers to the outer packaging that contains multiple units of the product or holds the primary packaging together. It serves various purposes, such as providing additional protection, facilitating transportation, and enhancing the presentation of the product. Here are some common considerations for secondary packaging requirements:

1. Product Protection: Secondary packaging should provide an extra layer of protection to the primary packaging and the product within. It should safeguard against physical damage, moisture, dust, and other potential hazards during handling, transportation, and storage.
2. Unitization and Grouping: Secondary packaging often involves grouping multiple units of the product together for convenience and efficiency. It should securely hold the individual units, preventing movement or shifting that could lead to damage.
3. Compatibility with Primary Packaging: Secondary packaging should be compatible with the size, shape, and design of the primary packaging. It should be able to accommodate the dimensions and configurations of the primary packaging units without compromising their integrity.
4. Labeling and Information Display: Secondary packaging should have sufficient surface area for labeling, including important product information, barcodes, logos, and branding elements. It should allow for clear visibility and legibility of the labels to facilitate easy identification and scanning.
5. Stacking and Palletization: Secondary packaging should be designed to allow for efficient stacking and palletization during transportation and storage. It should have structural integrity to support the weight of multiple units and withstand the pressure of stacking without collapsing or damaging the product.
6. Transportation and Handling: Secondary packaging should be designed to withstand the rigors of transportation, including impacts, vibrations, and potential rough handling. It should be durable enough to protect the product throughout the supply chain, from manufacturing to distribution to retail.
7. Sustainability and Environmental Considerations: Secondary packaging should take into account environmental sustainability by using recyclable or biodegradable materials whenever possible. It should align with relevant environmental standards and regulations.
8. Branding and Marketing: Secondary packaging provides an opportunity to enhance brand visibility and promote the product. It should be designed to reflect the brand's identity, communicate key messages, and attract consumer attention on store shelves.
9. Ease of Opening and Re-closing: Secondary packaging should be designed for easy opening by consumers while maintaining its integrity during transportation and storage. Consideration should also be given to re-closing features, if applicable, to allow for convenient resealing of the package.
10. Cost Efficiency and Manufacturing Considerations: Secondary packaging should be designed for cost-effective manufacturing and assembly processes. It should optimize material usage, minimize waste, and integrate well with automated packaging systems.

5.17. FOOD RECALL AND PRINCIPLES

Food recall refers to the process of removing and recalling food products from the market due to safety concerns or non-compliance with regulatory requirements. It is an important measure taken to protect consumers from potential health risks associated with contaminated or unsafe food.

Key principles and considerations for food recall:

1. Consumer Safety: The primary objective of a food recall is to protect consumer safety and minimize the risk of adverse health effects. The recall should be initiated promptly when there is evidence or reasonable suspicion that a food product poses a safety hazard.

2. Effective Communication: Clear and timely communication is crucial during a food recall. The responsible party, typically the manufacturer or distributor, should promptly notify relevant stakeholders, including regulatory agencies, retailers, and consumers. Communication channels should be established to reach consumers who may have purchased the affected product.
3. Traceability and Documentation: Maintaining accurate records and implementing effective traceability systems are essential for efficient and targeted recalls. Manufacturers should be able to identify the affected product batches or lots, track their distribution, and communicate this information to relevant parties.
4. Risk Assessment: A thorough risk assessment should be conducted to evaluate the potential hazards associated with the product and determine the appropriate level of recall. Factors such as the nature of the hazard, the likelihood of exposure, and the severity of potential health consequences should be considered.
5. Recall Strategy and Plan: A well-defined recall strategy and plan should be developed in advance to ensure an organized and effective recall process. The plan should outline roles and responsibilities, communication procedures, methods of retrieval, disposal or correction of affected products, and mechanisms for monitoring the effectiveness of the recall.
6. Regulatory Compliance: Food recalls should comply with applicable laws, regulations, and guidelines set by regulatory authorities. Close coordination with relevant regulatory agencies is crucial to ensure compliance and cooperation throughout the recall process.
7. Removal from Distribution: Affected products should be promptly removed from the distribution chain, including warehouses, retailers, and other points of sale. This may involve working closely with distributors, wholesalers, and retailers to retrieve the products.
8. Consumer Notification and Instructions: Clear instructions should be provided to consumers on how to identify the affected product, return or dispose of it, and seek any necessary medical attention if required. Communication should be conducted through various channels, including product labeling, websites, social media, and press releases.

5.18. PRODUCT STORAGE

Proper storage of vitamin A cassava products is essential to preserve their quality, prevent spoilage, and ensure food safety. Here are guidelines for storing vitamin A cassava products:

1. Temperature and Humidity: Store vitamin A cassava products in a cool, dry, and well-ventilated environment. Maintain a temperature of around 10 to 15 degrees Celsius (50 to 59 degrees Fahrenheit) to avoid moisture absorption, mold growth, and insect infestation.
2. Packaging: Choose food-grade packaging materials that provide protection against moisture, light, and contaminants. Ensure the packaging is properly sealed to maintain the quality and freshness of the cassava products.
3. Dedicated Storage Area: Designate a specific storage area for vitamin A cassava products. Keep the area clean, pest-free, and separate from other food items or non-food products to prevent cross-contamination.
4. FIFO (First-In, First-Out): Follow the FIFO principle to rotate the stored cassava products. Use the oldest stock first and place newly processed products behind the existing ones to ensure freshness and prevent product expiration.
5. Shelving and Stacking: Use sturdy shelves or pallets to store vitamin A cassava products. Keep the storage area organized, allowing proper air circulation between pallets or shelves. Avoid overstocking, which can lead to compression and damage.
6. Pest Control: Implement effective pest control measures to prevent insect infestation. Regularly inspect the storage area for signs of pests and take appropriate measures to control them.
7. Cleaning and Sanitation: Maintain cleanliness and sanitation in the storage area. Regularly clean shelves, floors, and storage containers to remove dust, debris, and spilled cassava products. Follow proper sanitation procedures using approved cleaning agents.

8. Monitoring and Inspection: Regularly monitor the stored vitamin A cassava products for spoilage, mold growth, or pest infestation. Conduct visual inspections and sensory evaluations to ensure product quality. Check packaging integrity and address any issues promptly.
9. Record-Keeping: Keep detailed records of the stored cassava products, including batch numbers, receipt dates, and expiration dates. This information aids in inventory management, rotation, and traceability.
10. Training and Education: Train employees on proper storage practices, including temperature control, product handling, and hygiene protocols. Educate them about the importance of maintaining product quality and recognizing signs of spoilage.

5.19. PRODUCT TRANSPORTATION AND DISTRIBUTION REQUIREMENTS

To ensure the quality and safety of cassava during transportation and distribution, the following recommendations should be followed:

1. Packaging: Use appropriate packaging materials that protect cassava from physical damage, moisture, pests, and contamination. Ensure clean, properly sealed bags made of durable materials like polypropylene or jute.
2. Handling and Loading: Train personnel on proper handling and loading procedures to avoid rough handling and damage to cassava. Maintain clean loading areas free from contaminants.
3. Transportation Vehicles: Choose clean, well-maintained vehicles suitable for transporting cassava. Ensure trucks or containers are dry, pest-free, and adequately ventilated. Regularly inspect and clean vehicles to maintain hygiene.
4. Temperature Control: Consider using refrigerated trucks or insulated containers for long-distance transportation to control temperature fluctuations. Preserve the quality and nutritional value of temperature-sensitive cassava.
5. Pest Control: Implement measures to prevent insect and pest infestation during transportation. Regularly inspect vehicles and ensure proper cleaning before loading cassava. Use insect-repellent materials or fumigation methods if necessary.
6. Transportation Duration: Minimize transportation time to limit exposure of cassava to adverse conditions. Plan efficient routes and schedules to reduce delays and maintain cassava freshness.
7. Documentation and Traceability: Maintain proper documentation and records throughout the transportation and distribution process. Include information on origin, quantity, quality, and handling of cassava to ensure traceability.
8. Collaboration with Logistics Partners: Work closely with logistics partners and transportation companies to ensure they understand cassava transportation requirements. Establish clear communication channels and provide guidelines on handling, storage, and transportation practices.
9. Compliance with Regulations: Adhere to local, national, and international regulations and standards for transporting agricultural products. Comply with food safety regulations, labeling requirements, and specific regulations for cassava products.
10. Quality Control and Monitoring: Regularly monitor cassava condition during transportation and distribution. Conduct quality checks at various stages to ensure compliance with required standards. Address any issues promptly to maintain cassava integrity.